

## US Economic Presentation

### US Economic Outlook 2026-2028: still bumpy



#### Economics

##### Americas

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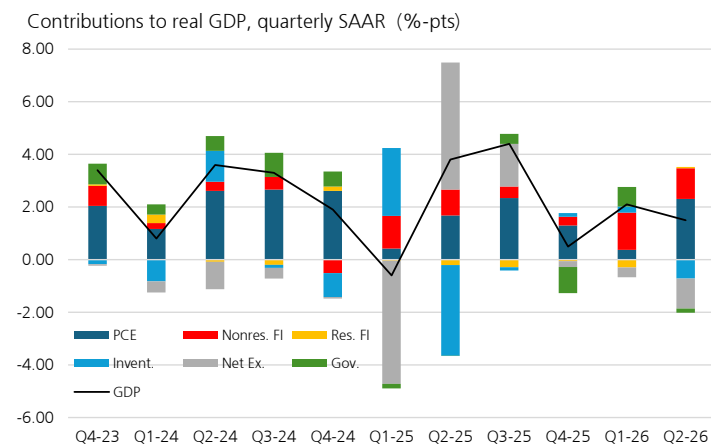
# US 2026 to 2028: Bumpy

- The expansion rests on AI. As the coming pages show, the economy's resilience in 2022 through 2024, with the insulation of strong fiscal support, has given way to the relentless AI/Tech capex and the wealth that has created, keeping spending growing faster than income.
- The rest of the economy does not look that strong. The rise in oil prices added another shock pushing inflation away from the FOMC's target. Chairman Warsh's Jackson Hole speech was unabashedly hawkish. We expect monetary policy to tighten with 0.25 pp rate hikes at the September and December FOMC meetings.
- OBBBA fiscal support is peaking. Fiscal policy flips to a drag in 2027, working in the same direction as monetary policy. That has not happened since both fiscal and monetary policy supported growth during the pandemic.
- Markets will need to adapt to less FOMC transparency, a new voice and reaction function. And more volatility. Our first note of 2026 said "*we expect a bumpy 2026.*" We thought growth would be ok, but not be smooth sailing.

| Variable           | 2023 | 2024 | 2025 | 2026 | 2027 | 2028 |
|--------------------|------|------|------|------|------|------|
| Change in real GDP | 3.4  | 2.4  | 2.0  | 2.2  | 1.9  | 2.6  |
| Unemployment rate  | 3.8  | 4.2  | 4.5  | 4.2  | 4.5  | 4.4  |
| PCE inflation      | 2.9  | 2.6  | 2.8  | 3.6  | 2.0  | 1.9  |
| Core PCE inflation | 3.3  | 3.0  | 2.8  | 3.1  | 2.4  | 2.1  |
| Federal funds rate | 5.4  | 4.4  | 3.6  | 4.1  | 4.1  | 3.6  |

Note: Q4/Q4 % basis except the unemployment rate which is a Q4 quarterly average. The federal funds rate reflects the mid-point of the target range. Years after 2025 are forecasts. Inflation reflects post-Sep. 30 NIPA revisions.

Source: BEA, BLS, Haver, UBS

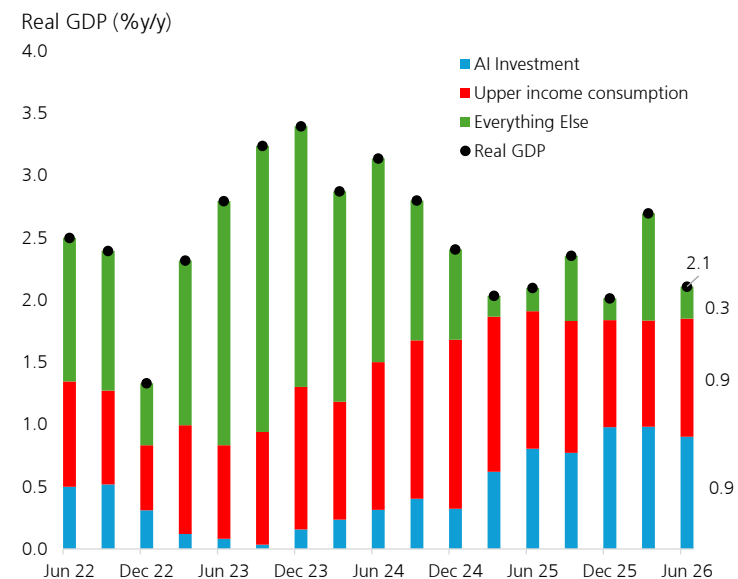


Source: BEA, Haver, UBS

# With AI driving the resilience and the only bright spots

*AI and tech are driving the strength in investment and likely supporting upper income households' consumption.*

- Over the last four quarters, structures and residential investment fell. Many households appear to be under pressure. Government spending has flatlined. That's a lot of the economy contracting or stagnating.
- Equity market wealth represents essentially all the increase in real wealth the past several years and hit an all-time high share of household net worth, 38%, in Q2.
- We expect AI to drive a regime shift to higher structural productivity growth. We expect disruption, including labor market disruption, to rise alongside that shift.
- The economy's reliance on AI and AI-driven wealth could be a vulnerability too. The energy price shock is a risk to firm costs and spending. Geopolitical volatility may be a headwind. Headline resilience masks risks.

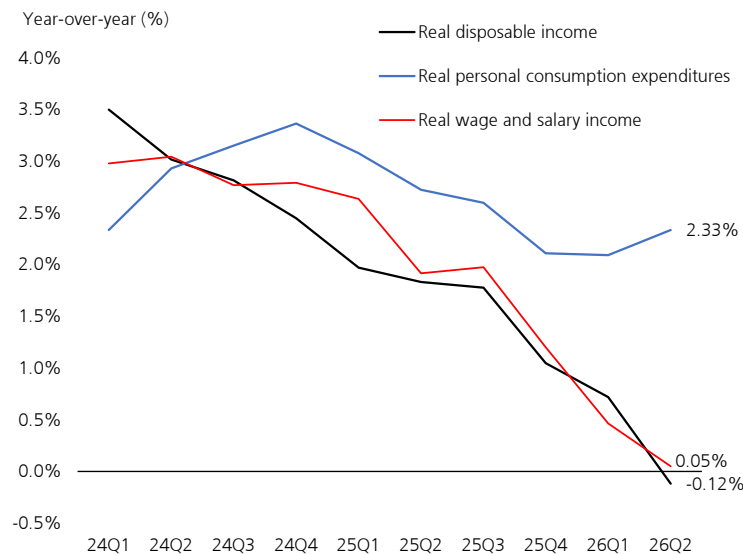


Source: BEA, Haver, UBS

# Households: bifurcated...tariffs, energy prices and refunds

*Tariffs and energy prices are eroding the boost from OBBBA. The labor market slowdown has lowered income growth. Real spending growth continues to handily outpace income, as AI wealth supports financing spending from saving.*

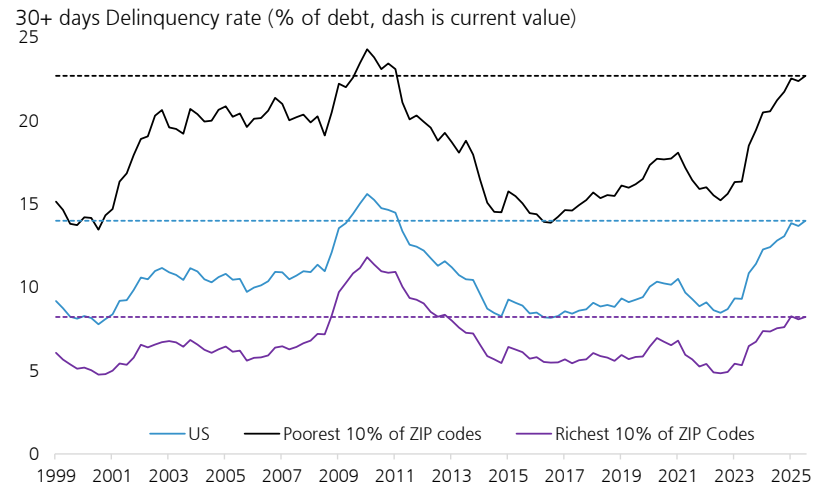
Consumer spending holding up as income growth slows



Source: BEA, Haver, UBS

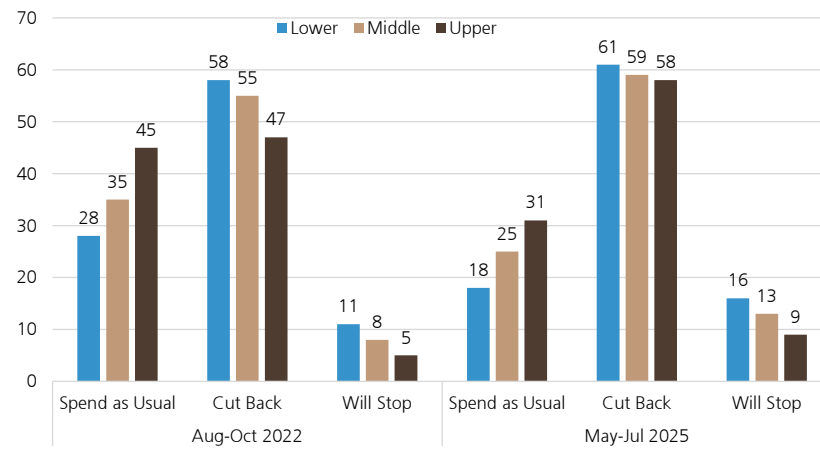
- The US consumer continues to spend. Spending has slowed on net from the robust pace of 2024, but not nearly as much as real disposable income growth.
- Households across all income groups expect to be more sensitive to future inflation now than in 2022. At the same time, households are less able to weather inflation. Households below the top 20% of the income distribution have relatively low levels of liquid assets.
- Equity market wealth in Q2 was an all-time high share of household wealth (38%) and has accounted for all of the gains in wealth the last few years. The S&P 500 helps.
- We have seen a ~\$60 billion boost to tax refunds, thanks to the OBBBA, so we should be passing peak fiscal support for spending. The key question beyond that:

*Can spending keep outpacing income to this degree?*

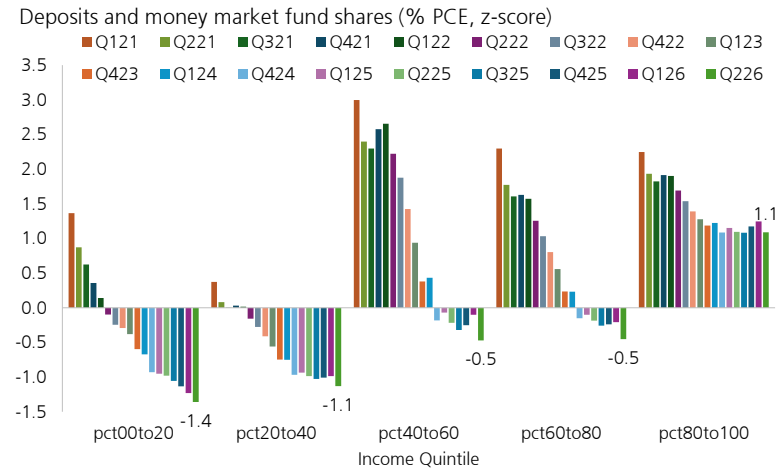


Source: FRB St Louis, Haver, UBS

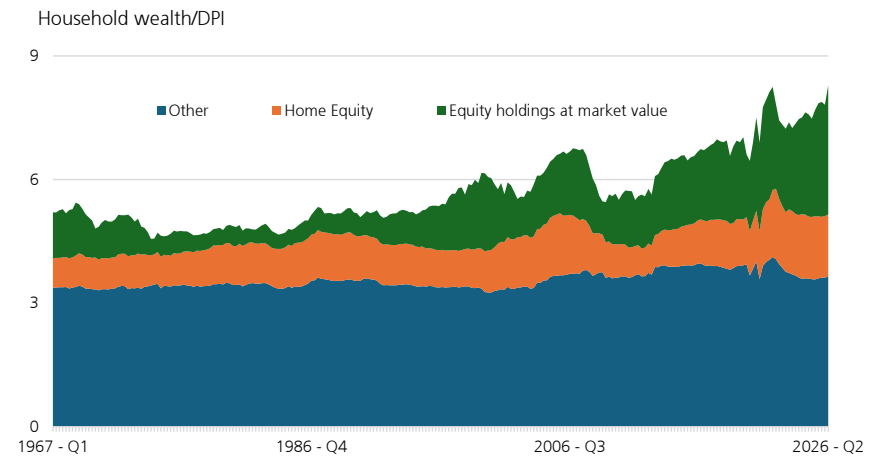
Year-ahead anticipated spending response to inflation (by income group)



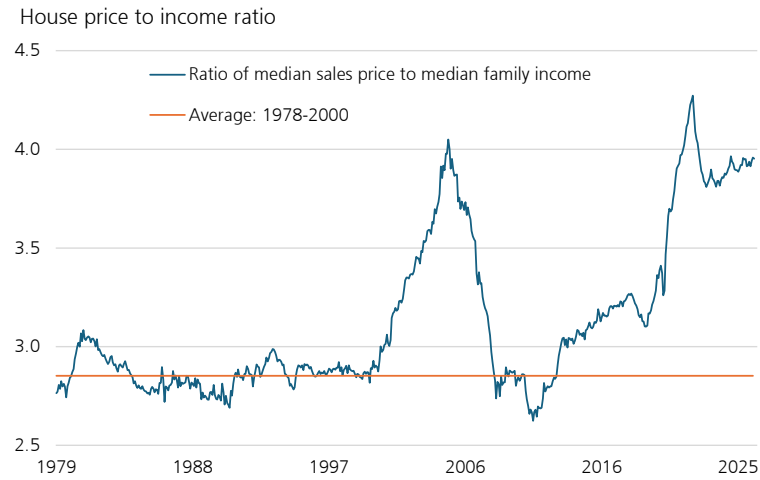
Source: University of Michigan, UBS



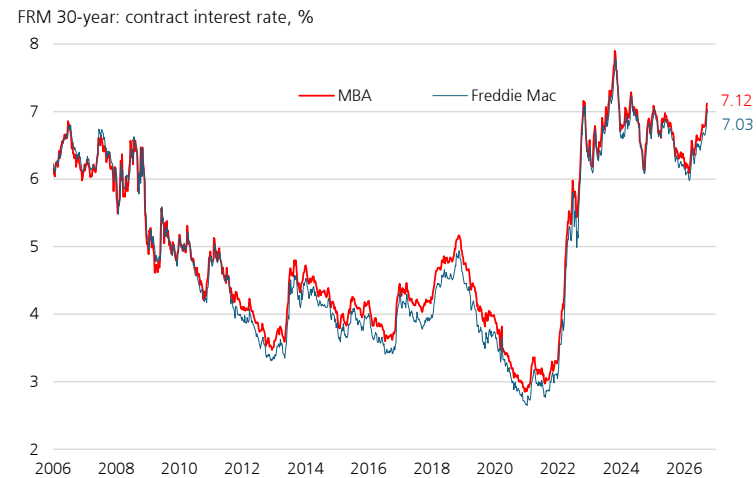
Source: Federal Reserve, Haver, UBS



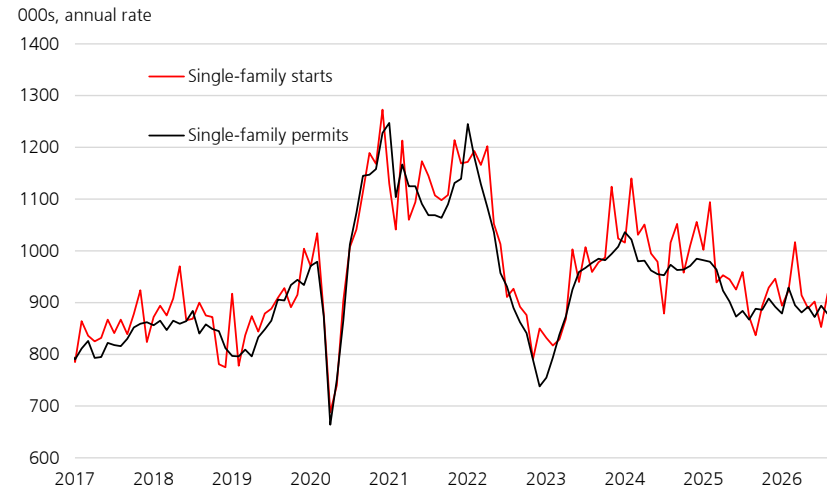
Source: Federal Reserve, Haver, UBS



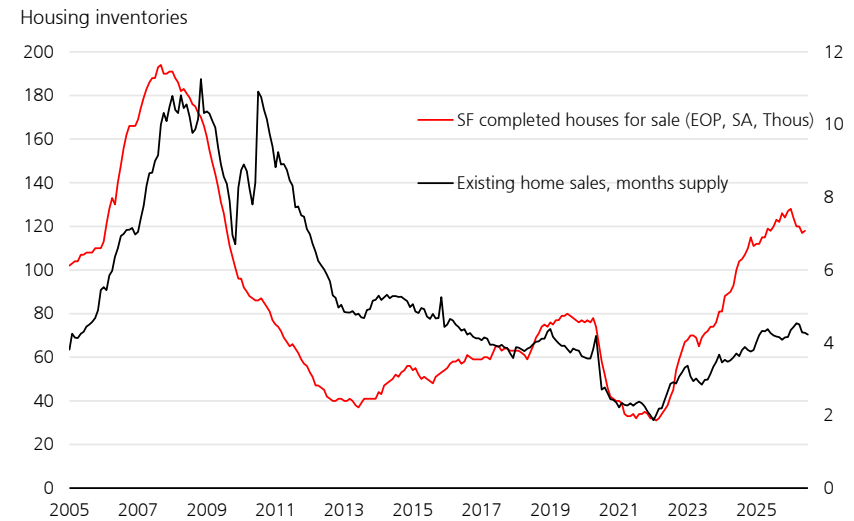
Source: National Association of Realtors, Census, Haver, UBS



Source: Mortgage Bankers Association, Haver, UBS



Source: Census Bureau, Haver, UBS

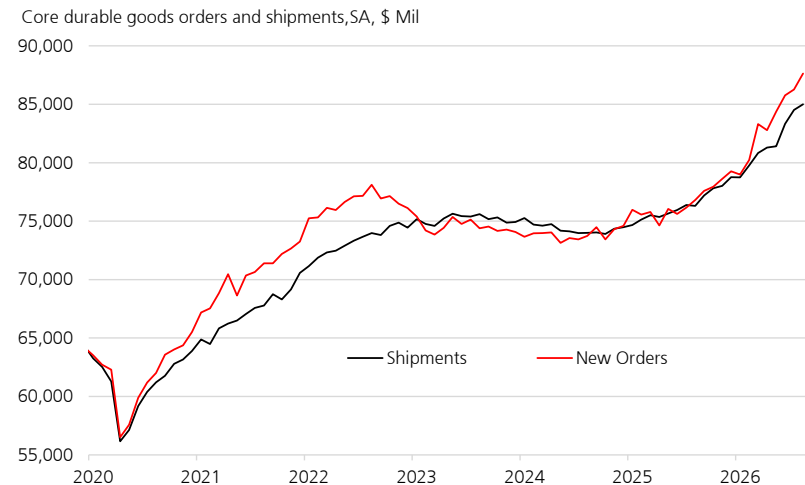


Source: Census Bureau, Haver, UBS

# Business sector: AI and nothing else?

*AI-related investment is the bright spot, everything else doesn't look that hot.*

- Business fixed investment growth is almost all AI. Over the past 4 quarters, the AI/tech-related real equipment investment increased a robust 20% (table next page). The rest of equipment investment fell -2%. Within IP, software rose 11%, R&D 8%, and the rest fell -0.6%.
- Other US investment has been weak, with some broadening though most of that has been due to AI too, boosting demand for everything from generators to ceramics. Real non-residential structures and residential investment have contracted for two years.
- We expect OBBBA is supporting investment now too, and some broadening beyond AI. Some inventory rebuilding is underway. But fiscal support wanes from here, restocking may help for a while, but AI/tech capex will likely remain the dominant theme.
- In sum, a large share of investment has been in recession through Q2. Inventory alignment might help. Now rates look headed higher too.



Source: Census Bureau, Haver, UBS

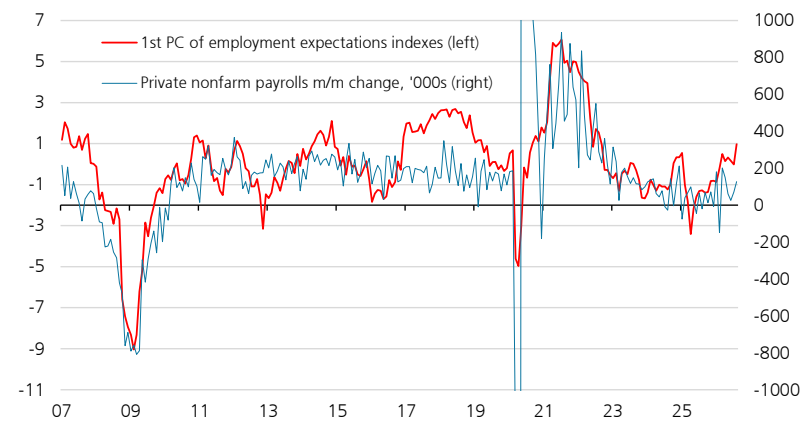
# Investment looks weak except for tech/AI...

*Tech/AI strong. The rest not so much. The bottom row shows the 4-quarter changes. Non-AI falling in real terms.*

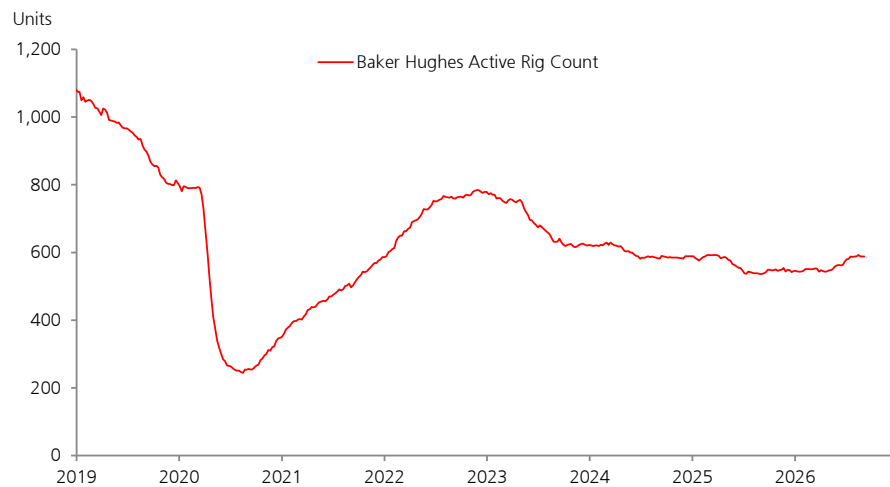
| Real \$bn | <u>Structures investment</u> |                    | <u>IP investment</u> |                  |                 | Real \$bn | <u>Equipment investment</u> |                    |                     |                    |
|-----------|------------------------------|--------------------|----------------------|------------------|-----------------|-----------|-----------------------------|--------------------|---------------------|--------------------|
|           | <u>Non-residential</u>       | <u>Residential</u> | <u>R&amp;D</u>       | <u>Ent/other</u> | <u>Software</u> |           | <u>Non-AI</u>               | <u>Pot. AI sum</u> | <u>Info process</u> | <u>Other/elect</u> |
| 2024 - Q1 | 704.8                        | 796.2              | 666.9                | 92.7             | 763.0           | 2024 - Q1 | 693.1                       | 606.9              | 498.3               | 108.7              |
| 2024 - Q2 | 697.8                        | 792.2              | 664.4                | 92.3             | 769.9           | 2024 - Q2 | 709.5                       | 618.7              | 511.3               | 107.4              |
| 2024 - Q3 | 693.9                        | 782.5              | 669.5                | 92.3             | 774.7           | 2024 - Q3 | 715.4                       | 639.0              | 529.4               | 109.6              |
| 2024 - Q4 | 679.5                        | 790.7              | 663.4                | 92.4             | 780.1           | 2024 - Q4 | 696.8                       | 643.4              | 530.2               | 113.2              |
| 2025 - Q1 | 674.2                        | 788.8              | 660.2                | 91.5             | 814.3           | 2025 - Q1 | 696.4                       | 709.6              | 598.8               | 110.9              |
| 2025 - Q2 | 661.2                        | 778.5              | 673.7                | 90.2             | 863.8           | 2025 - Q2 | 698.8                       | 735.5              | 615.6               | 119.9              |
| 2025 - Q3 | 652.8                        | 764.3              | 687.5                | 89.8             | 871.5           | 2025 - Q3 | 704.6                       | 747.9              | 628.7               | 119.2              |
| 2025 - Q4 | 641.8                        | 761.0              | 699.5                | 89.0             | 881.7           | 2025 - Q4 | 668.4                       | 799.3              | 680.2               | 119.2              |
| 2026 - Q1 | 634.2                        | 745.7              | 712.3                | 89.5             | 929.1           | 2026 - Q1 | 662.0                       | 860.2              | 739.7               | 120.5              |
| 2026 - Q2 | 631.3                        | 748.0              | 723.9                | 89.7             | 956.8           | 2026 - Q2 | 688.2                       | 883.1              | 752.1               | 131.0              |
| Changes:  |                              |                    |                      |                  |                 | Changes:  |                             |                    |                     |                    |
| d(4Q)     | -29.9                        | -30.5              | 50.2                 | -0.5             | 93.0            | d(4Q)     | -10.6                       | 147.6              | 136.5               | 11.1               |
| d(2Q)     | -10.5                        | -13.0              | 24.4                 | 0.7              | 75.1            | d(2Q)     | 19.8                        | 83.8               | 71.9                | 11.9               |
| %(4Q)     | -4.5                         | -3.9               | 7.5                  | -0.6             | 10.8            | %(4Q)     | -1.5                        | 20.1               | 22.2                | 9.3                |

Source: BEA, Haver, UBS

Employment expectations

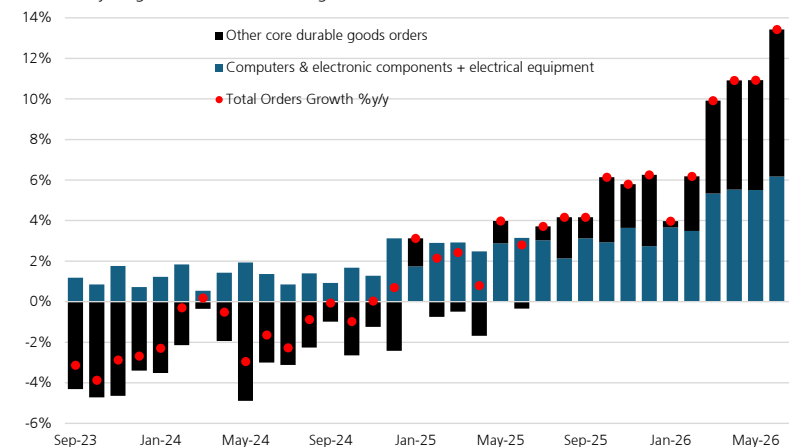


Source: BEA, Haver, UBS



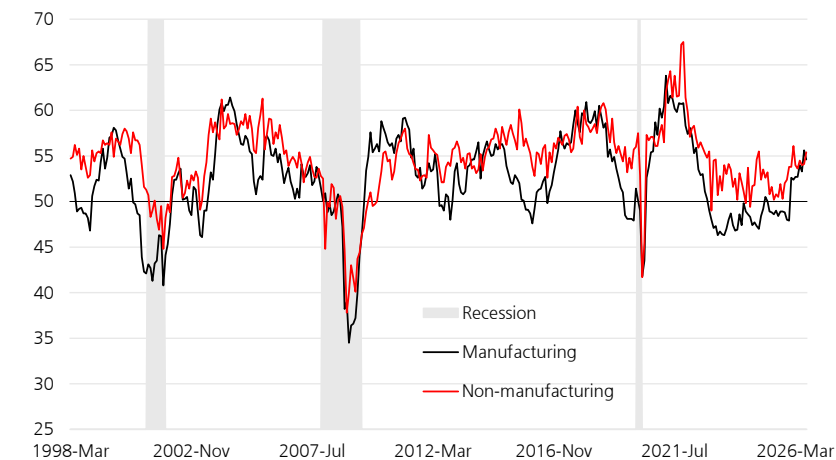
Source: Baker Hughes, Haver, UBS

Year-over-year growth in core durable goods orders



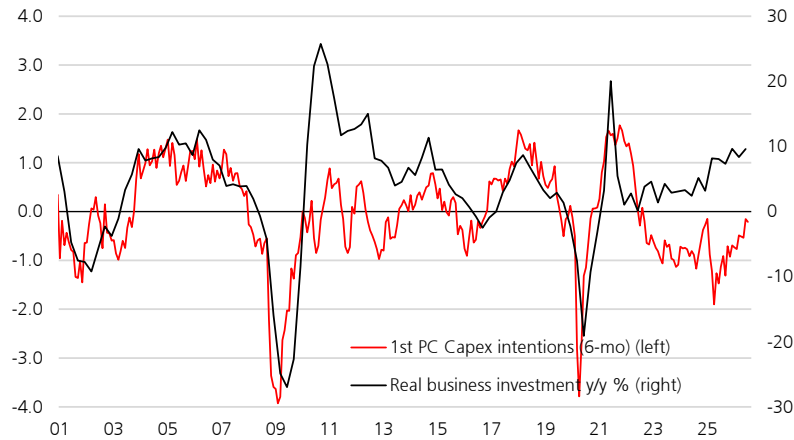
Source: Census Bureau, Haver, UBS

ISM indexes



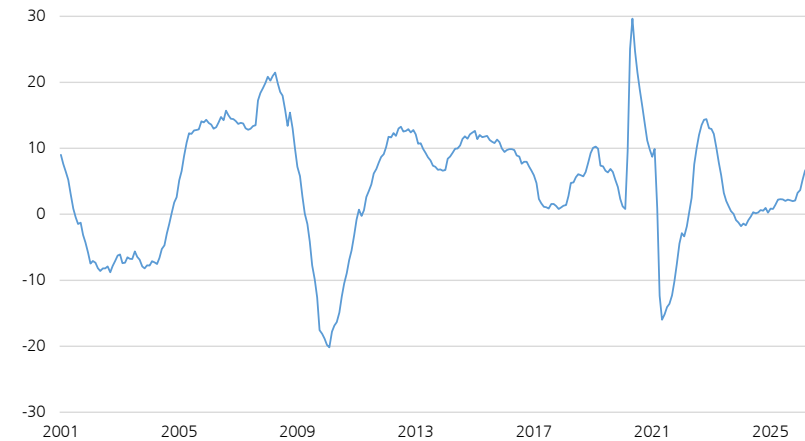
Source: ISM, Haver, UBS

Capex intentions versus business investment



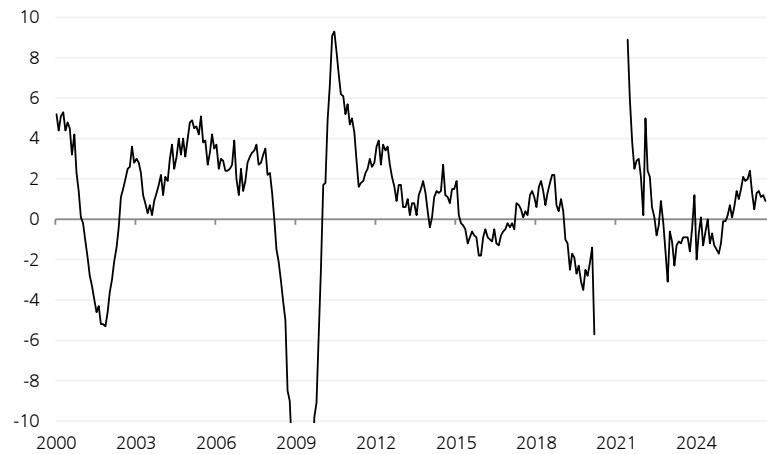
Source: Federal Reserve System, NFIB, NABE, Business Roundtable, Haver, UBS

C&I lending, break-adjusted  
Year-to-year pct. chg.



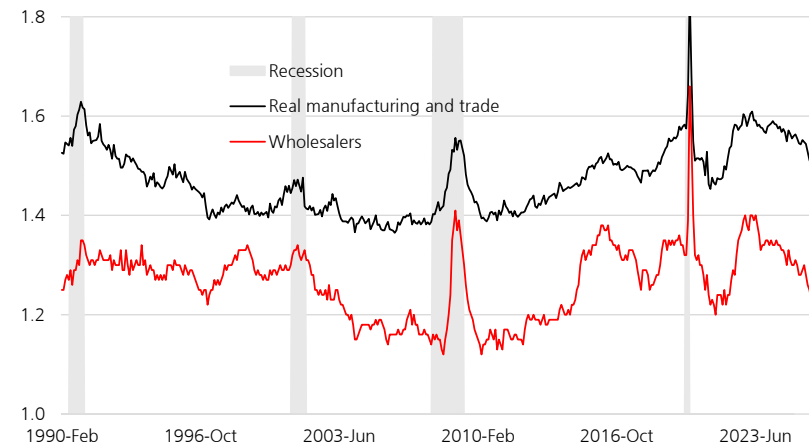
Source: Federal Reserve, Haver, UBS

Manufacturing production (April 2020 to May 2021 omitted)  
Y/Y % chg.



Source: Federal Reserve, Haver, UBS

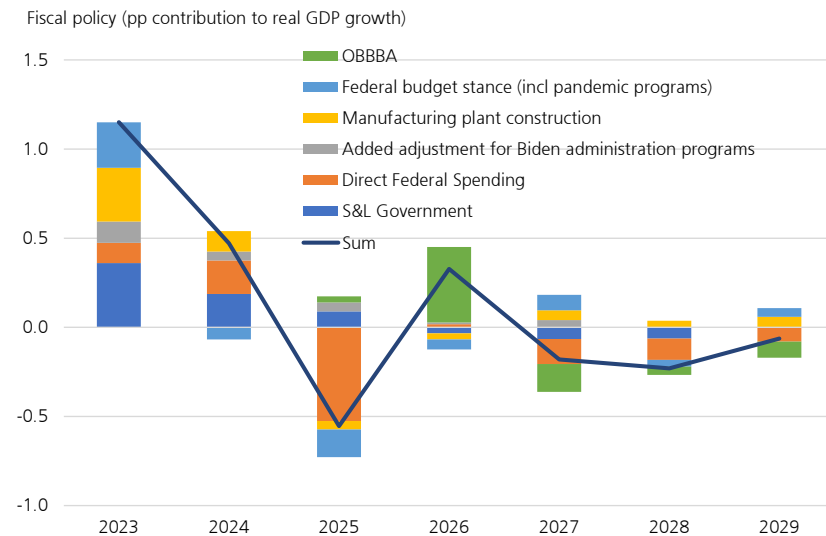
Inventory/sales ratios



Source: Census, Haver, UBS

# Government: One Big Beautiful Bill Act support to wane

*Fiscal to support to 2026 growth with front-loaded OBBBA tax cuts, should fade with cuts in years 2027 through 2029.*

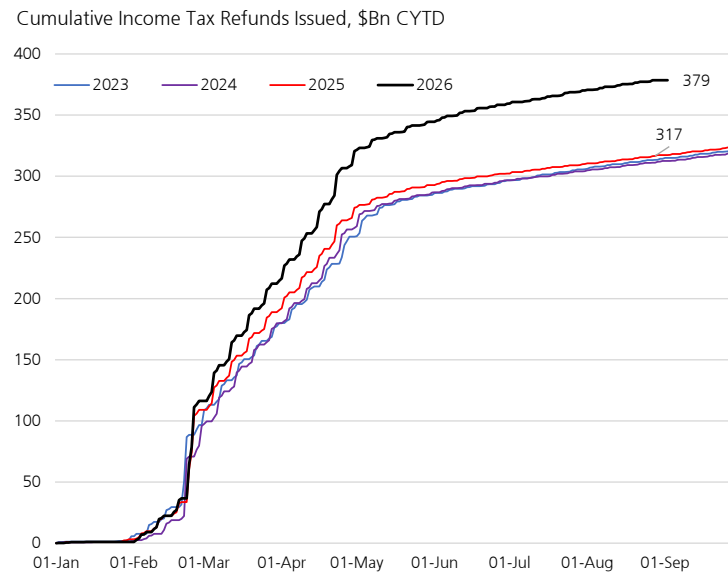


Source: CBO, JCT, BEA, Census, Haver, UBS

- We estimate fiscal policy contributed 1.1 pp to growth in 2023 and 0.5 pp to 2024 before turning contractionary in 2025. OBBBA flipped that to support 2026. In 2027, budget cuts turn the fiscal impulse negative again after the mid-term elections.
- Beyond the current ~\$95 billion defense/farm bill, we do not expect meaningful fiscal bills out of Congress. They still need to replace the 5-year transportation funding bill expiring, pass a defense authorization, and avoid a September 30<sup>th</sup> government shutdown.
- Even with new tariff revenue of ~\$250 billion the next two years, the federal deficit is likely to remain a little over 6% of GDP in our view.
- State and local governments are under pressure too, and federal cuts are pushing costs back to them.

# Tax refunds flowing to households

*OBBBA tax refunds added \$60 billion to households in H1 2026. More generous expensing provisions for business fixed investment lowered 2026 corporate taxes. That and other OBBBA provisions added to the fiscal impulse in 2026.*

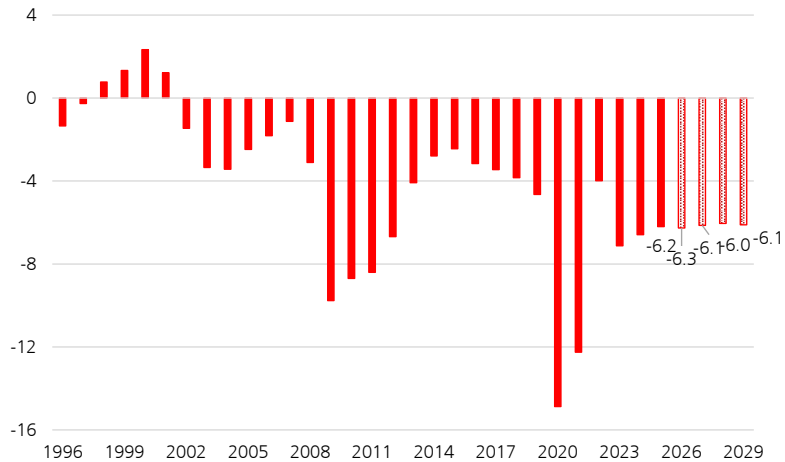


Source: U.S. Treasury, Haver, UBS

| Deficit widening (FY \$bns)            |      |      |      |      |      |
|----------------------------------------|------|------|------|------|------|
| Proposal (deficit widening (FY \$bns)) | 2025 | 2026 | 2027 | 2028 | 2029 |
| Tax policies:                          |      |      |      |      |      |
| Extend TCJA individual provisions      | -7   | -210 | -437 | -427 | -425 |
| Child tax credit expansion             | 0    | -9   | -5   | -7   | -8   |
| Deductions for seniors, tips, overtime | -11  | -71  | -64  | -64  | -33  |
| Expand TCJA business provisions        | -100 | -129 | -100 | -82  | -63  |
| Other individual                       | -13  | -55  | -51  | -54  | -56  |
| Other tax cuts                         | 0    | 0    | 0    | -9   | -7   |
| Repeal IRA tax credits                 | 0    | 9    | 34   | 49   | 54   |
| Other tax hikes                        | 0    | 5    | 28   | 0    | 0    |
| Spending cuts:                         |      |      |      |      |      |
| Medicaid changes                       | 1    | 22   | 54   | 73   | 106  |
| ACA Changes                            | 0    | 5    | 13   | 17   | 18   |
| Other healthcare changes               | 0    | -1   | -1   | -1   | -5   |
| SNAP changes                           | 0    | 8    | 16   | 22   | 22   |
| Student loan programs                  | 150  | 12   | 12   | 13   | 13   |
| Other spending cuts                    | 22   | 4    | 12   | 17   | 13   |
| Spending increases                     |      |      |      |      |      |
| Defense                                | -2   | -42  | -45  | -25  | -17  |
| Border security + judiciary            | 0    | -12  | -21  | -31  | -33  |
| Other spending increases               | 0    | -13  | -21  | -10  | 0    |
| Deficit widening (current law)         | 40   | -479 | -588 | -535 | -434 |
| Deficit widening (current policy)      | 47   | -271 | -151 | -109 | -9   |

Source: CBO, JCT, UBS

Budget deficit adjusted for student loan accounting (FY, % of GDP)



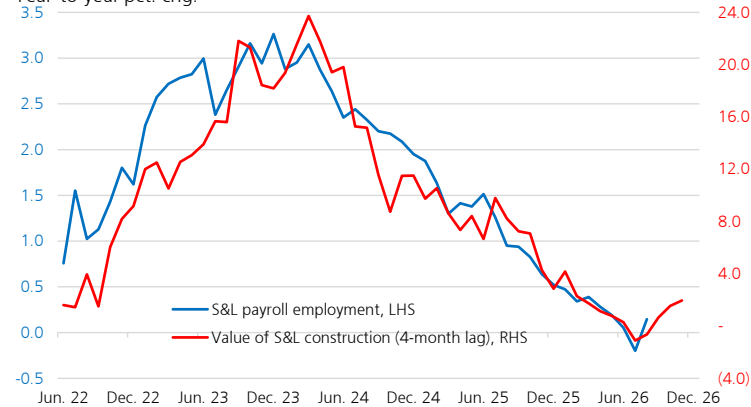
Source: CBO, Haver, UBS

Debt Held by the Public as a Percentage of GDP (Fiscal Yr, %)



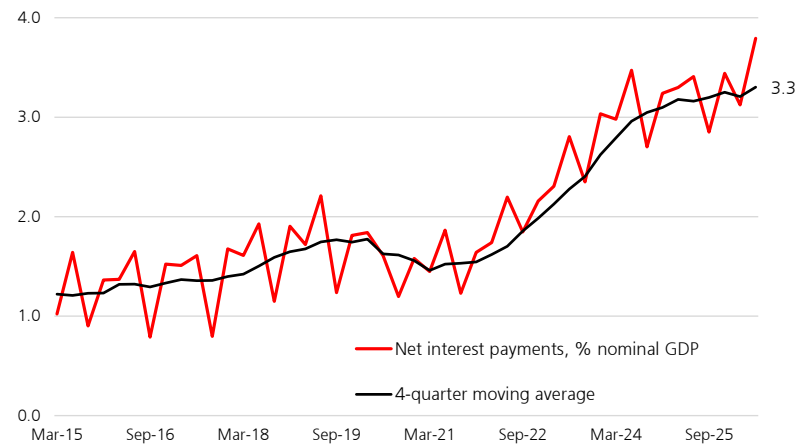
Source: U.S. Treasury, Haver, UBS

State & local government hiring and construction growth  
Year-to-year pct. chg.



Source: BLS, Census, Haver, UBS

Net interest payments as % of GDP

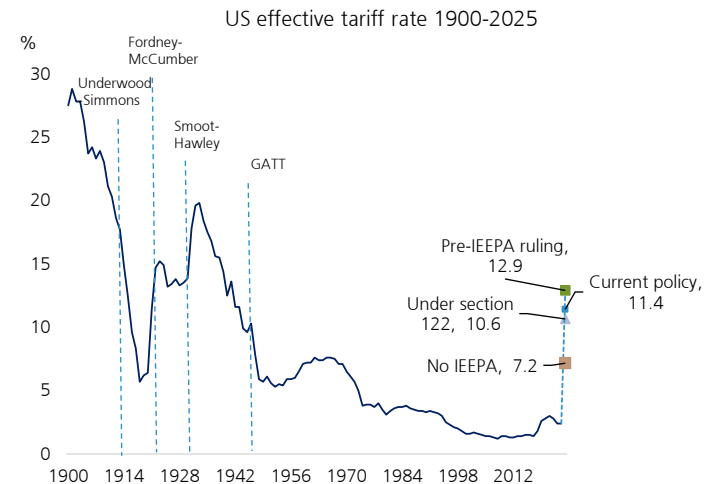


Source: U.S. Treasury, Haver, UBS

# Tariff policy headwinds remain

*The administration appears to have replicated most of the tariffs prior to the Supreme Court decision. However, implementation details and court cases could keep trade policy uncertainty elevated.*

- On July 24, the administration imposed new tariffs in which 19 trading partners face a 10% tariff rate and 41 face a 12.5% tariff rate.
- The latest authority, section 301 of the Trade Act of 1974, has withstood legal challenge, but the new actions have already been challenged in the courts due to the breadth of the action far beyond prior use of the statute. New tariffs came into effect on imports from Canada under section 338 of the 1930s Tariff Act. Also, potentially open to legal challenge. With tensions elevated between the two. Overall, we estimate the weighted average tariff rate is ~11.4%, edging higher with recent tariff actions.
- A refund process for ~\$160bn paid in IEEPA tariffs is now established. Refunds are flowing but mainly to larger importers, who are less likely to spend them.



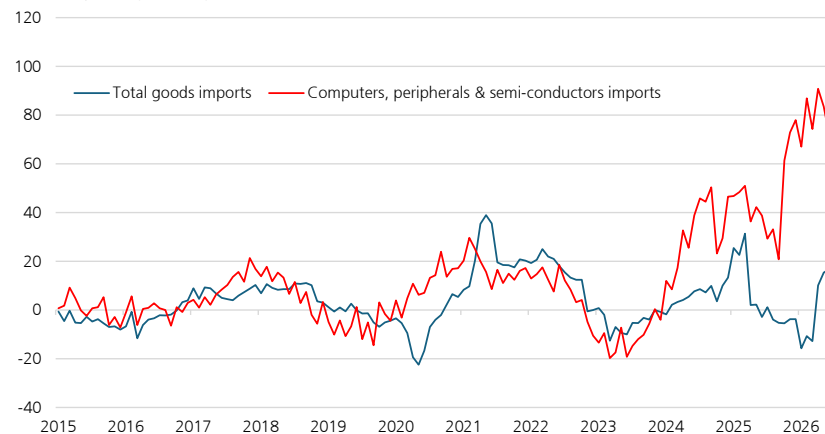
Source: Census, USATRADE, Haver, UBS

# Imports boosted by tech demand despite tariffs

*Imports supported by tech investment and trade diversion despite expectations for a tariff driven pull back in trade.*

- Tariff policy uncertainty could cause lumpiness in trade flows again this year. However, the administration walked back the high levels of tariffs in the April 2, 2025 announcement, mitigating the economic damage and decoupling.
- The economy is likely still adjusting to the impacts of higher tariffs, but imports have also been boosted by inputs into the AI buildout, with much of this tech imported. Import sources have shifted too, with imports from China declining, and imports from other Asian economies getting a boost.

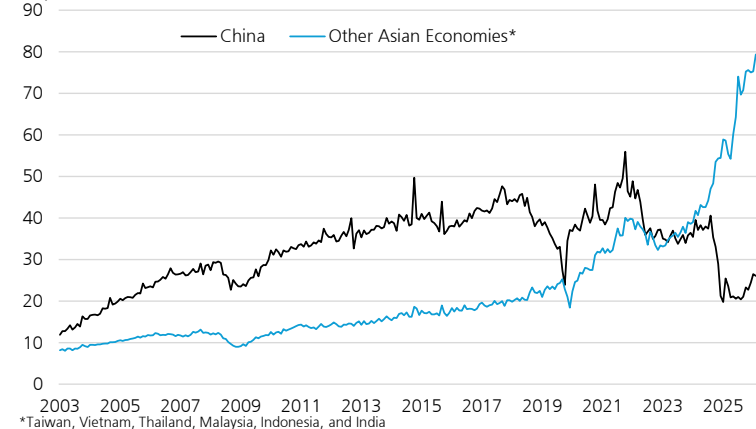
Goods imports (year-to-year %)



Source: Census, USATRADE, Haver, UBS



Imports (\$bn.)

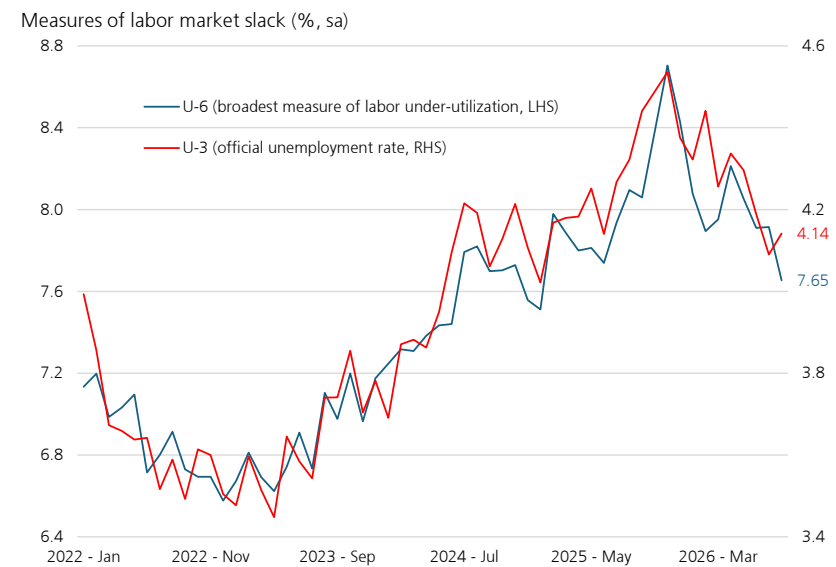


\*Taiwan, Vietnam, Thailand, Malaysia, Indonesia, and India

# Labor market: stable but not great

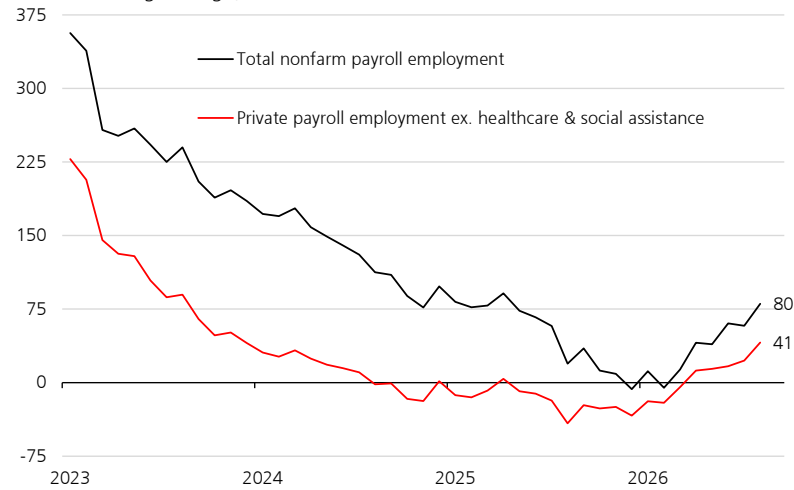
*The labor market still looks soft. Across data sources demand seems to have slowed as much as supply.*

- Private sector labor demand remains subdued. Excluding health care and social assistance, private nonfarm payroll employment has been adding 22K jobs per month for the 12 months ending in August, and work week look low..
- Household survey job growth underlying the unemployment rate has been very weak, but LFPR has dropped to help. Layoffs are not that low, but normal.
- Residual seasonality, little net immigration, and subdued labor demand likely remain in 2026. Some strength has cooled in H2, on net. Several sectors, like manufacturing carved out a bottom. Another help to job gains, the federal government likely won't shed 300K jobs in 2026.
- Still nominal wage growth has growth has slowed notably, suggesting the labor market may be slacker than the unemployment rate alone suggests.



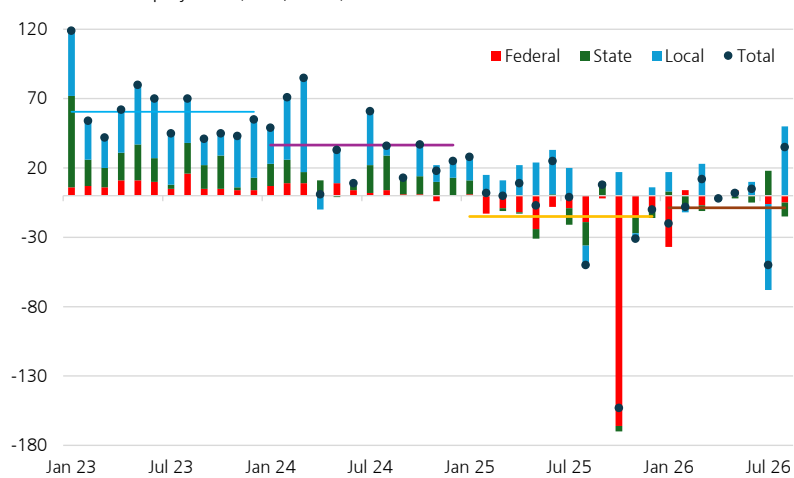
Source: BLS, Haver, UBS

8-month moving average, SA



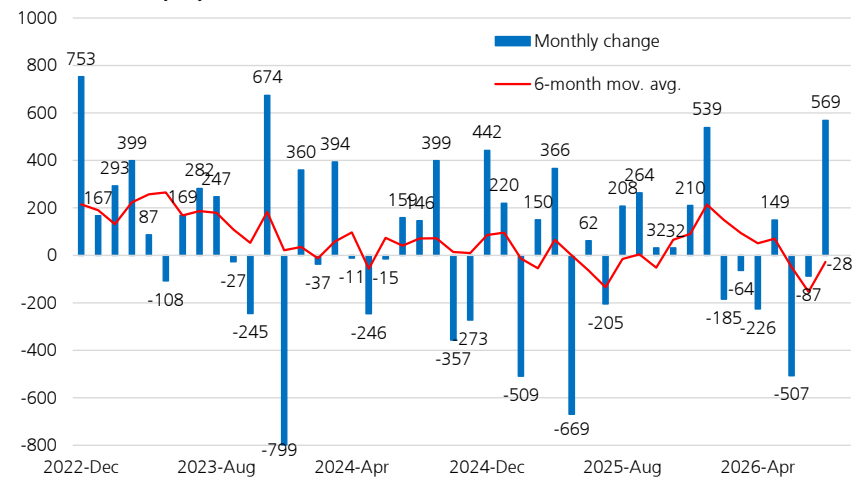
Source: BLS, Haver, UBS

Government employment (m/m, '000s)



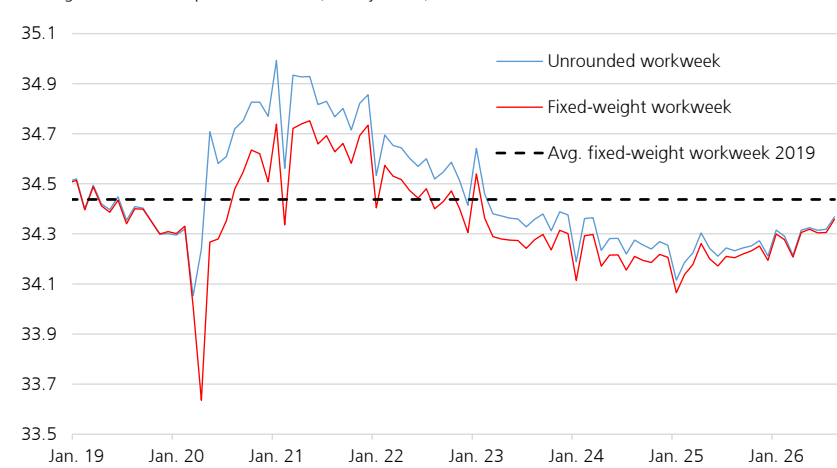
Source: BLS, Haver, UBS

Change in household survey employment, population-adjusted  
Thous., seasonally adjusted

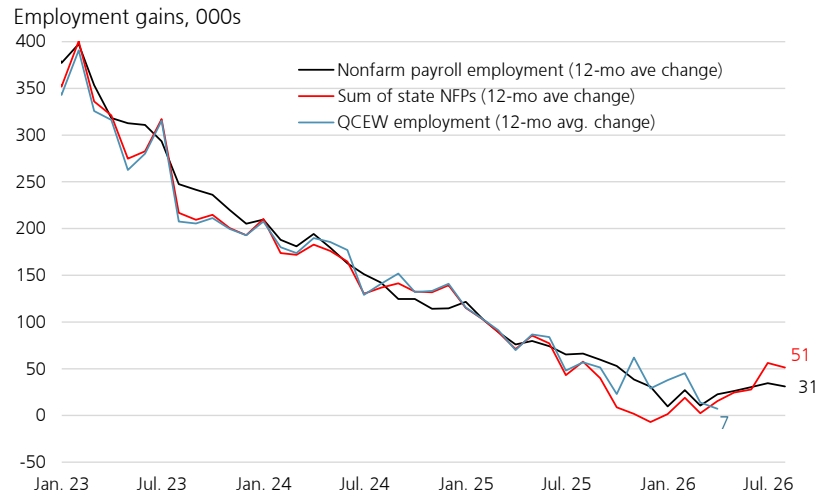


Source: BLS, Haver, UBS

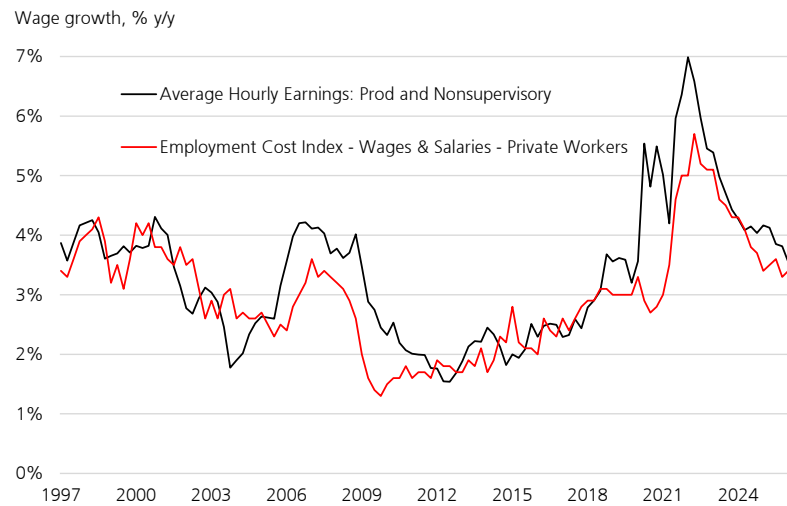
Average workweek of private workers (weekly hours)



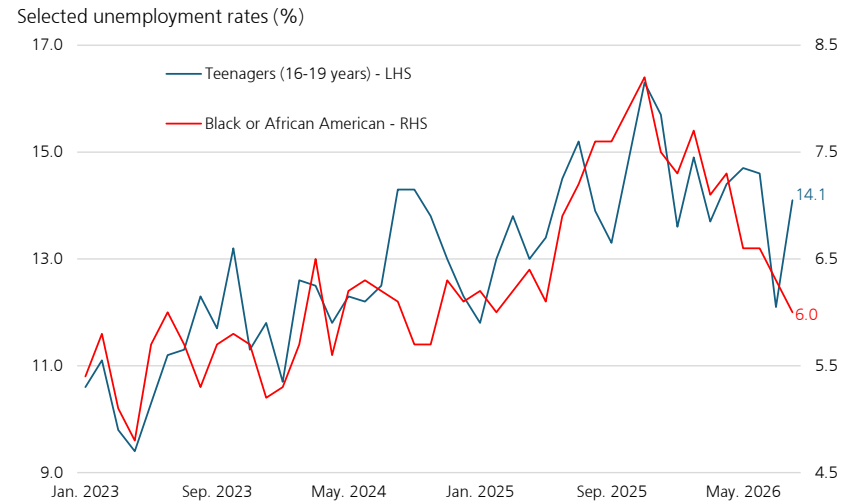
Source: BLS, Haver, UBS



Source: BLS, Haver, UBS

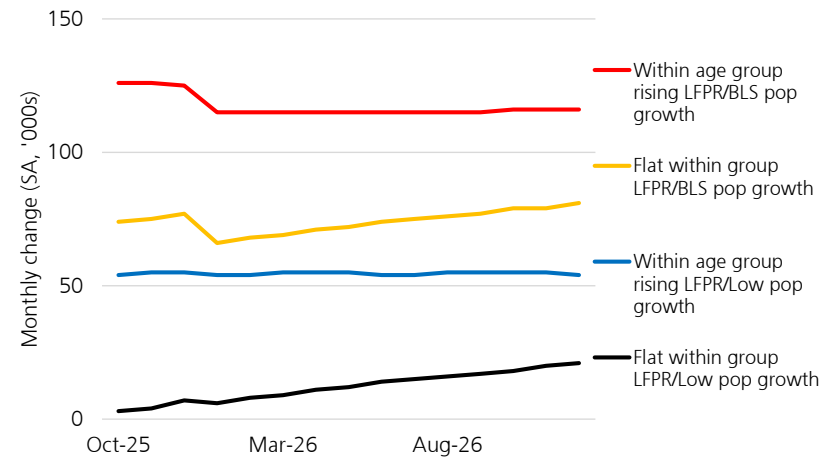


Source: BLS, Haver, UBS



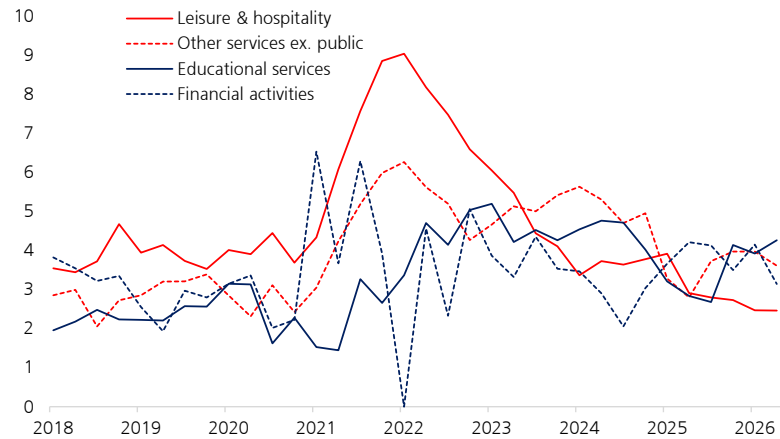
Source: BLS, Haver, UBS

Break-even: HH survey job gains that keep u-rate steady



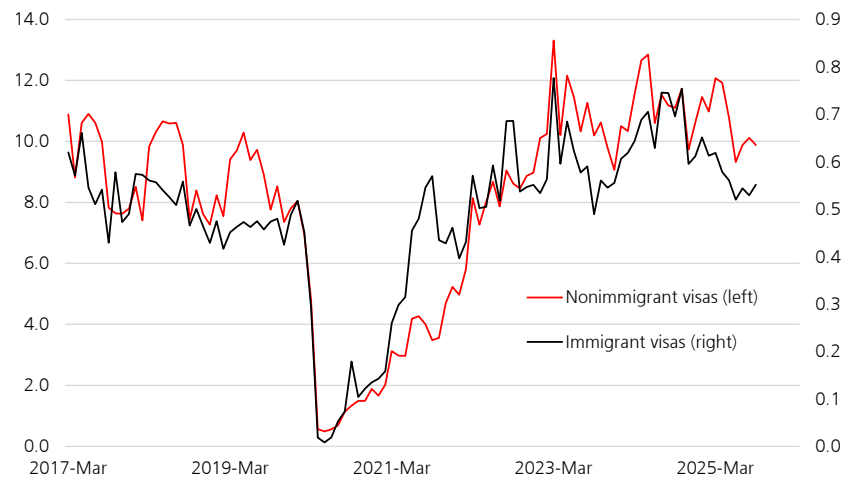
Source: Census, Haver, UBS

Employee cost index, private wages & salaries  
Year-to-year % chg.



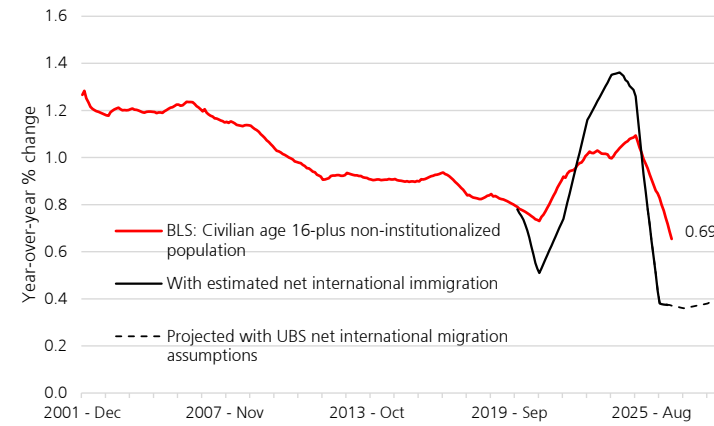
Source: BLS, Haver, UBS

Number of visas approved  
Mil., annualized, both scales



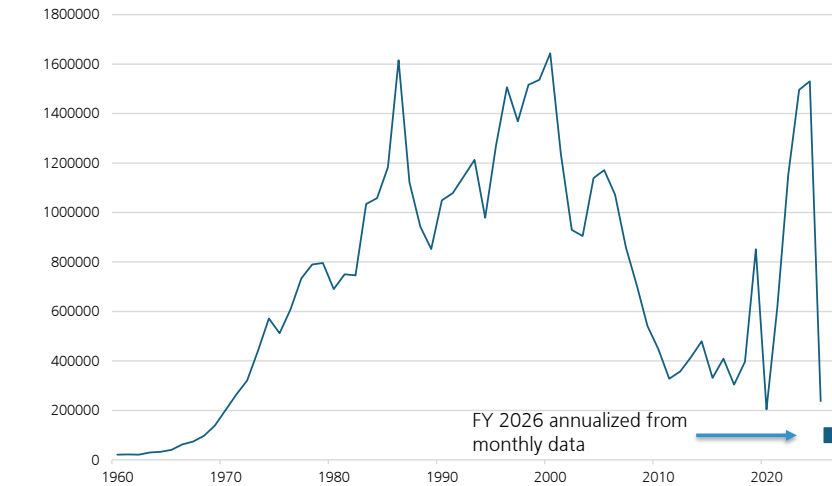
Source: US Department of State, Haver, UBS

HH Survey population growth  
(break-adjusted for population updates)

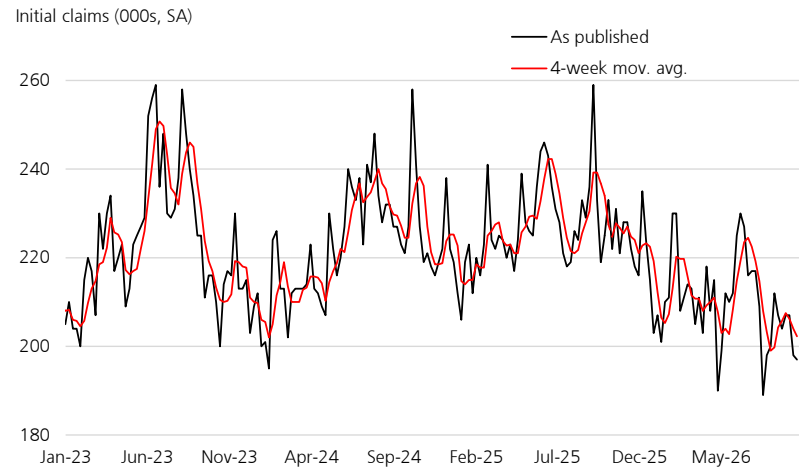


Source: BLS, Haver, UBS

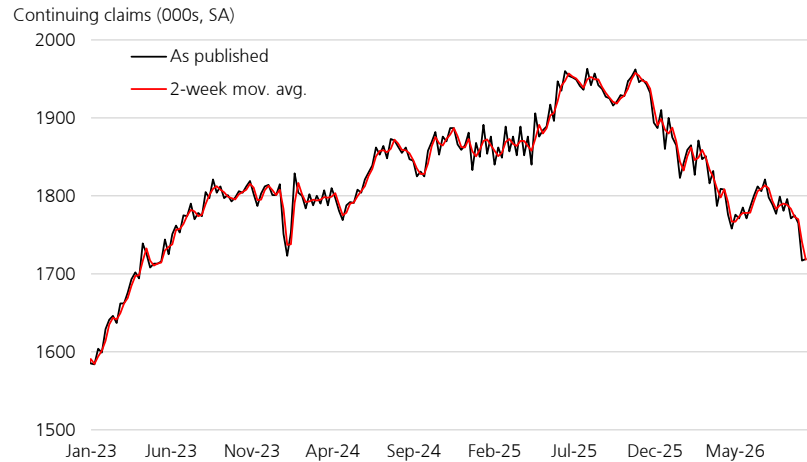
Southwest Land Border Apprehensions (FY, Number)



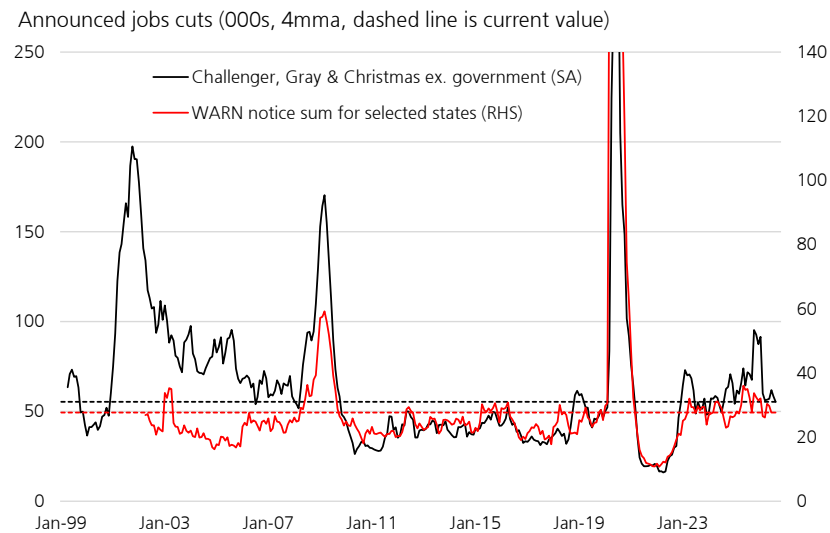
Source: Customs and Border Protection, Haver, UBS



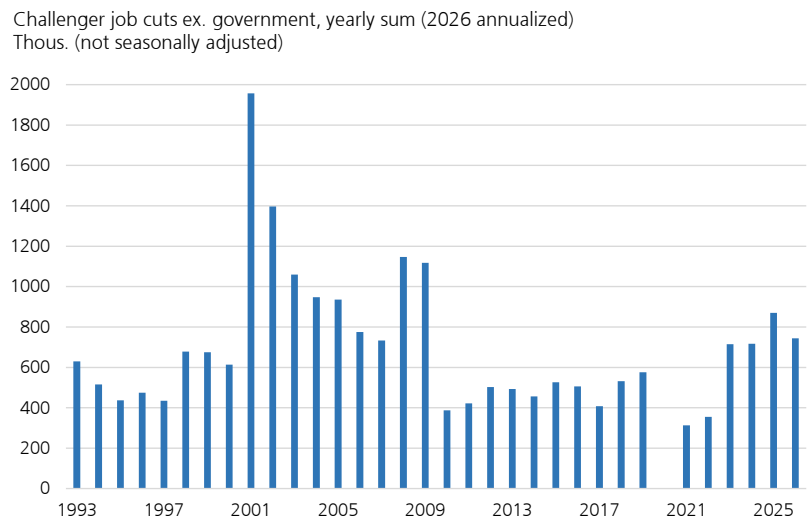
Source: DOL, Haver, UBS



Source: DOL, Haver, UBS



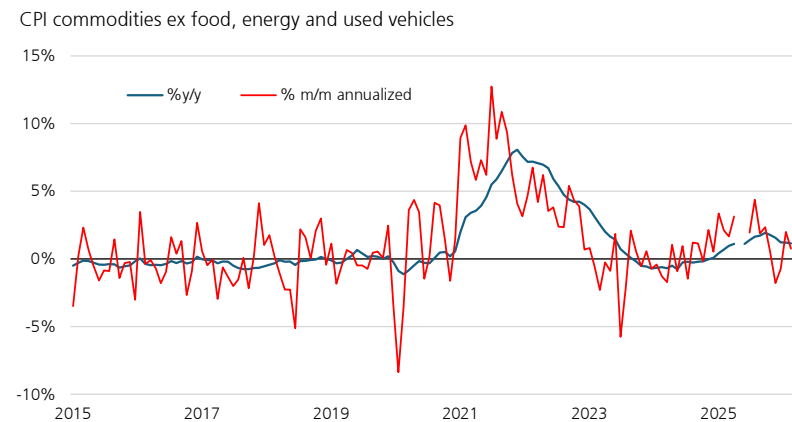
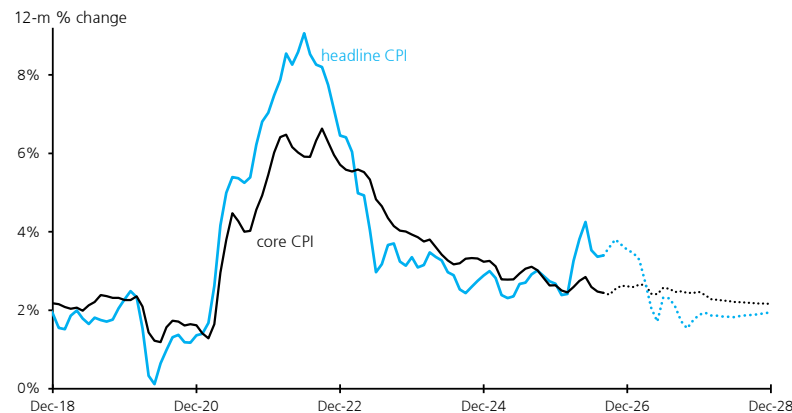
Source: Challenger, Gray & Christmas, Haver, UBS



Source: Challenger, Gray & Christmas, Haver, UBS

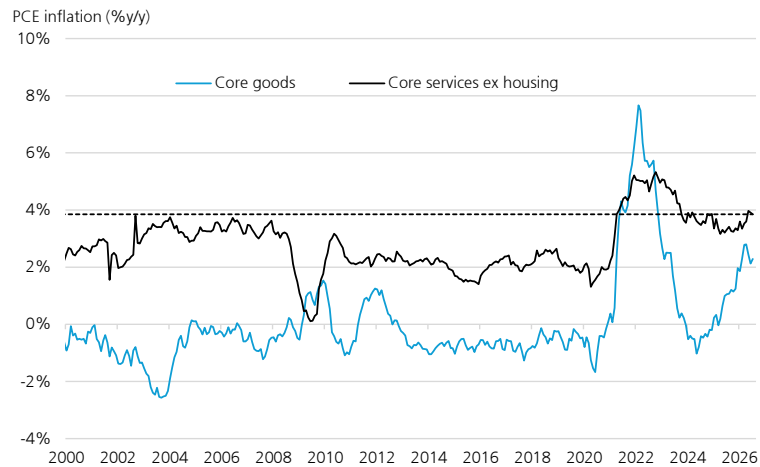
# Inflation: tariffs, energy prices, and AI

*Tariff price passthrough should have peaked. The energy price shock should wane, though AI is adding pressure. How high oil prices rise, how long they remain elevated, and what other supply is disrupted remains very uncertain.*

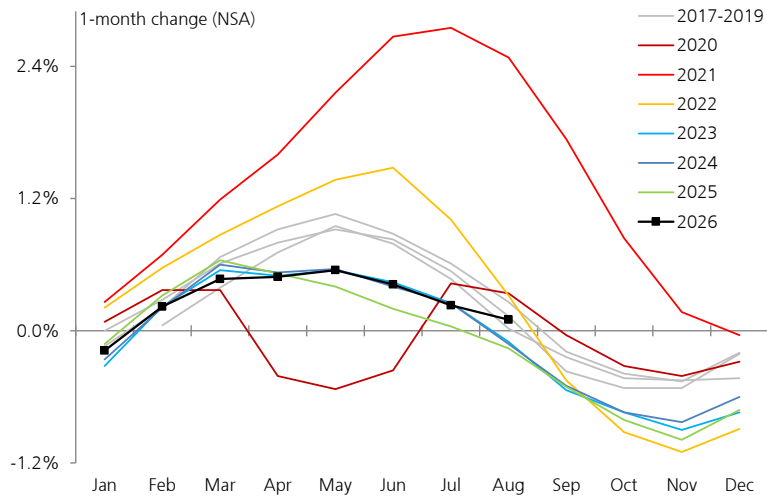


Source: BLS, BEA, Haver, UBS

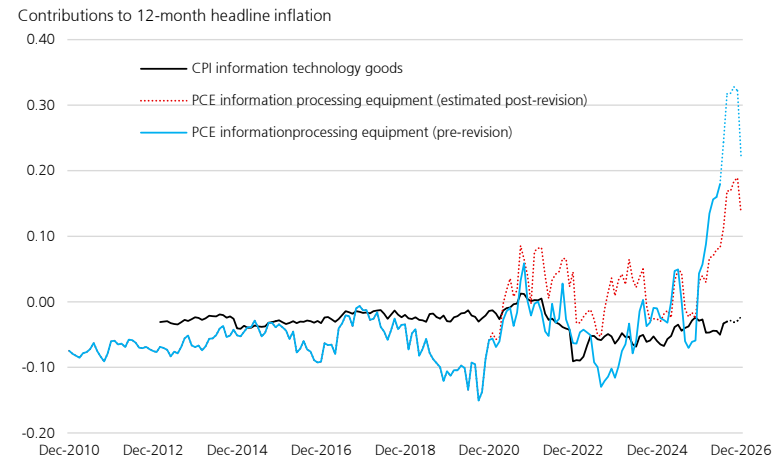
- By April 2025 core PCE inflation had fallen from the peak of 5.61% to 2.61% and headline inflation was down to 2.28%. Pretty good progress for the FOMC. That brought “liberation day” and widespread tariffs. This year energy prices have moved higher. The supply shocks are pushing inflation away from the 2.0% target.
- Tariff pass-through should be slowing but remain in coming quarters. Energy price increases may have only modest pass through to core, but not zero. Once the levels stop going up the growth rate falls. But, we do not know when the strait will open. Then, what next?
- We project core PCE inflation peaked at ~3.4% (year over year) in the May data. We expect inflation ebbs slowly with core PCE inflation at 3.1% for 2026 (Q4/Q4), after accounting for the BEA methodological adjustments and moderating more than 0.5 pp to around 2.4% in 2027... above target over five years.



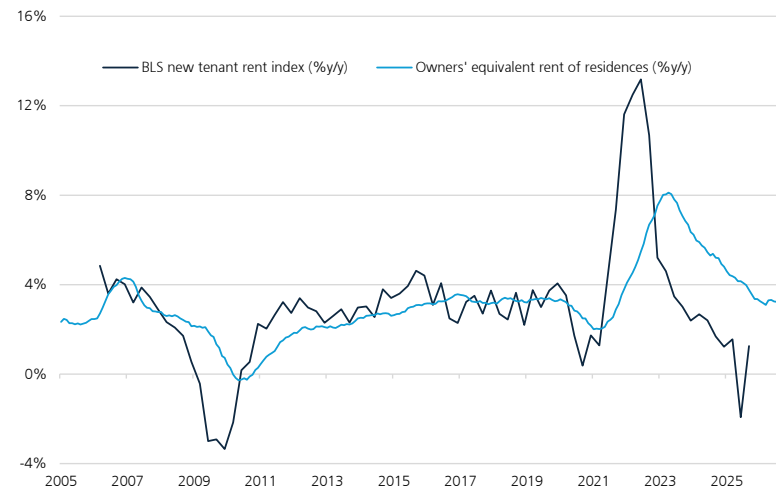
Source: BEA, Haver, UBS



Source: ApartmentList, Haver, UBS



Source: BLS, BEA, Haver, UBS



Source: BLS, Haver, UBS

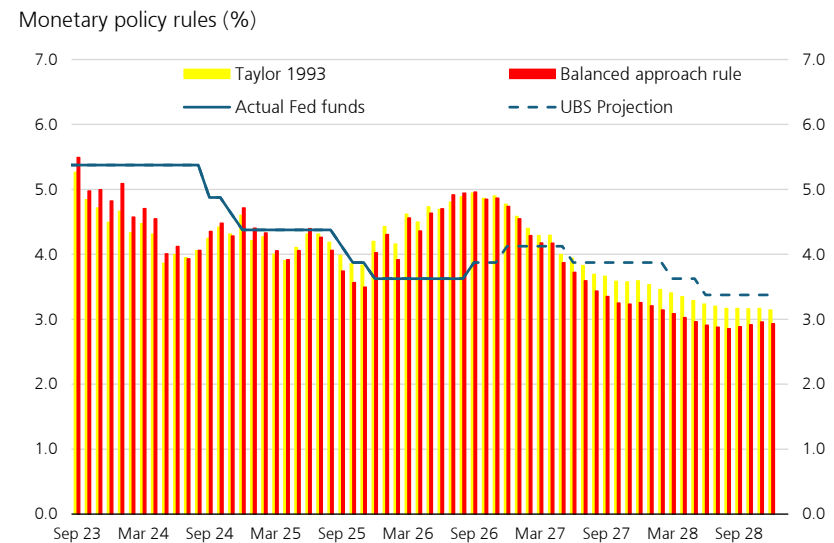
| Date   | PCE M/M (%) | Core PCE M/M | PCE Y/Y | Core PCE Y/Y |
|--------|-------------|--------------|---------|--------------|
| Jan-25 | 0.329       | 0.291        | 2.521   | 2.683        |
| Feb-25 | 0.379       | 0.422        | 2.567   | 2.806        |
| Mar-25 | 0.088       | 0.179        | 2.331   | 2.639        |
| Apr-25 | 0.197       | 0.223        | 2.259   | 2.594        |
| May-25 | 0.215       | 0.266        | 2.442   | 2.765        |
| Jun-25 | 0.256       | 0.228        | 2.605   | 2.820        |
| Jul-25 | 0.166       | 0.239        | 2.644   | 2.907        |
| Aug-25 | 0.225       | 0.183        | 2.712   | 2.872        |
| Sep-25 | 0.292       | 0.225        | 2.779   | 2.815        |
| Oct-25 | 0.213       | 0.251        | 2.718   | 2.761        |
| Nov-25 | 0.178       | 0.131        | 2.776   | 2.779        |
| Dec-25 | 0.253       | 0.240        | 2.829   | 2.916        |
| Jan-26 | 0.312       | 0.405        | 2.811   | 3.033        |
| Feb-26 | 0.338       | 0.325        | 2.769   | 2.933        |
| Mar-26 | 0.649       | 0.276        | 3.345   | 3.032        |
| Apr-26 | 0.360       | 0.198        | 3.512   | 3.006        |
| May-26 | 0.509       | 0.391        | 3.816   | 3.135        |
| Jun-26 | -0.069      | 0.173        | 3.480   | 3.078        |
| Jul-26 | 0.158       | 0.250        | 3.471   | 3.089        |
| Aug-26 | 0.315       | 0.267        | 3.565   | 3.176        |
| Sep-26 | 0.346       | 0.166        | 3.620   | 3.116        |
| Oct-26 | 0.292       | 0.221        | 3.702   | 3.085        |
| Nov-26 | 0.122       | 0.164        | 3.643   | 3.120        |
| Dec-26 | 0.173       | 0.159        | 3.560   | 3.036        |
| Jan-27 | 0.185       | 0.284        | 3.430   | 2.912        |
| Feb-27 | 0.153       | 0.255        | 3.238   | 2.840        |
| Mar-27 | 0.235       | 0.207        | 2.813   | 2.769        |
| Apr-27 | 0.113       | 0.204        | 2.560   | 2.775        |
| May-27 | 0.160       | 0.201        | 2.204   | 2.581        |
| Jun-27 | 0.183       | 0.196        | 2.462   | 2.604        |
| Jul-27 | 0.091       | 0.191        | 2.393   | 2.543        |
| Aug-27 | 0.176       | 0.191        | 2.251   | 2.466        |
| Sep-27 | 0.158       | 0.185        | 2.059   | 2.485        |
| Oct-27 | 0.150       | 0.182        | 1.914   | 2.445        |
| Nov-27 | 0.191       | 0.164        | 1.984   | 2.445        |
| Dec-27 | 0.221       | 0.163        | 2.033   | 2.449        |

\*Based on post-BEA revision data.

Source: BEA, Haver, UBS

# FOMC is in a tough spot: hike to slow what?

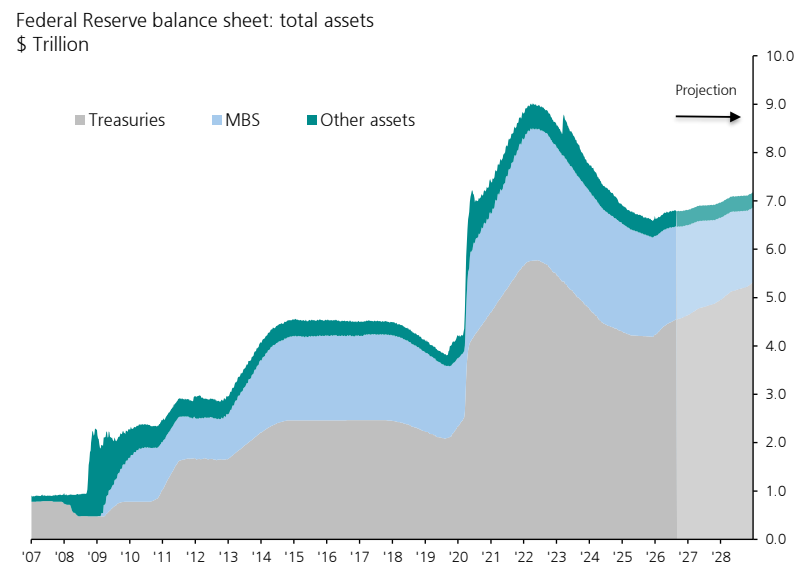
- The FOMC is in a tough spot. The funds rate cannot do much to stop cost-push inflation. Domestic sectors like the labor market, housing, and households that pay interest are not inflationary, and slowing them further may not offset the supply shocks.
- Markets will need to adjust to new Chairman Warsh's communications: less transparency, clarity, and guidance; more risk of mispricing and volatility.
- Having hiked once in September, we project the FOMC hikes one more time this year in December. We view the distribution of potential outcomes for the funds rate over the next two years as quite wide, in both directions, hikes or cuts.
- After Jackson Hole, Chair Warsh clearly put rate hikes on the table and doubled down on this in September post-meeting press conference. We expect another hike in December this year. Market pricing will likely matter, geopolitical developments, as well as underlying inflation improvement.



# Reserve management purchases to continue

*Reserve management purchases began in 2025 at ~\$40bn per month, this stepped down to \$25bn, then \$10bn.*

- Reserve management purchases began December 12. The FRBNY reduced the pace of expected purchases to \$25 billion per month as of mid-April and then lowered that further to \$10 billion per month, has now stopped purchases.
- We assume that pace picks back up at some point, and penciled in that the monthly pace of purchases will be lifted to ~\$30 billion per month in 2027 in order to ensure ample reserves as the Fed's other liabilities rise.
- There is a task force to consider changes in balance sheet policy and composition. FOMC members and Chair Kevin Warsh have signaled potential broader changes to the balance sheet but only slowly implemented. We expect that other than regulatory changes underway, consideration of issues will continue but progress slowly and balance sheet expansion generally continues.

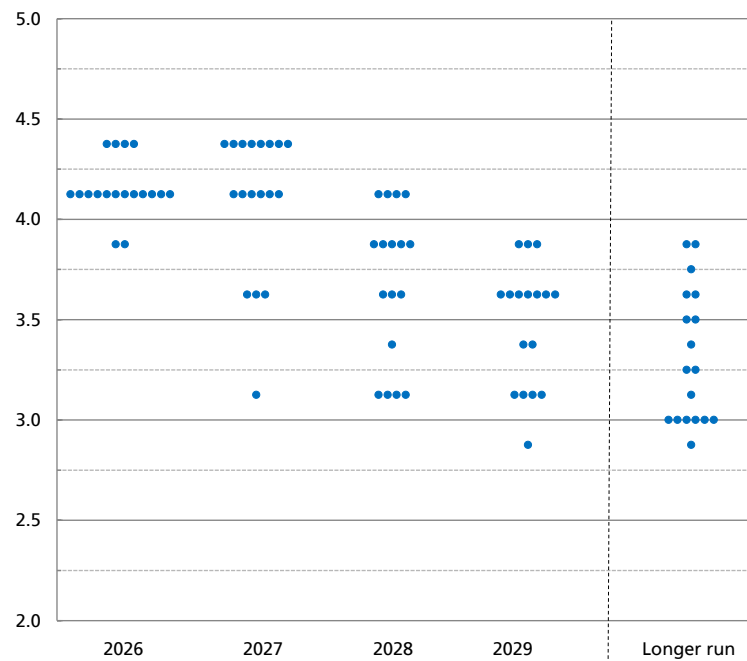


Source: Federal Reserve, Haver, UBS

# Personnel is policy: the new Chair is one of 19

*A new Fed chair, the ongoing legal case into the removal of Governor Cook, and the potential for further personnel change add uncertainty to the outlook. Markets will need to adjust to changes in things like communication style too.*

FOMC participants' assessments of appropriate monetary policy: midpoint of target range or target level for the federal funds rate, September 2026 (pct.)



Source: Federal Reserve, Haver, UBS

- Who is Chairman Warsh? We are still finding out. What we are increasingly learning is that he is not like his predecessors. He has staked out an even more hawkish reaction function. Framing Chairman Warsh's FOMC, past templates may no longer apply. This increasingly looks like a new regime, and a bigger break from the past than appreciated.

FOMC September Summary of Economic Projections (medians)

| <u>Variable:</u>   | <u>2026</u> | <u>2027</u> | <u>2028</u> | <u>2029</u> | <u>LR</u> |
|--------------------|-------------|-------------|-------------|-------------|-----------|
| Change in real GDP | 2.3         | 2.4         | 2.2         | 2.1         | 2.0       |
| June projection    | 2.2         | 2.3         | 2.2         |             | 2.0       |
| Unemployment rate  | 4.1         | 4.1         | 4.1         | 4.1         | 4.2       |
| June projection    | 4.3         | 4.3         | 4.2         |             | 4.2       |
| PCE inflation      | 3.7         | 2.3         | 2.1         | 2.0         | 2.0       |
| June projection    | 3.6         | 2.3         | 2.0         |             | 2.0       |
| Core PCE inflation | 3.4         | 2.5         | 2.2         | 2.0         |           |
| June projection    | 3.3         | 2.5         | 2.1         |             |           |
| Federal funds rate | 4.1         | 4.1         | 3.9         | 3.6         | 3.2       |
| June projection    | 3.8         | 3.6         | 3.4         |             | 3.1       |

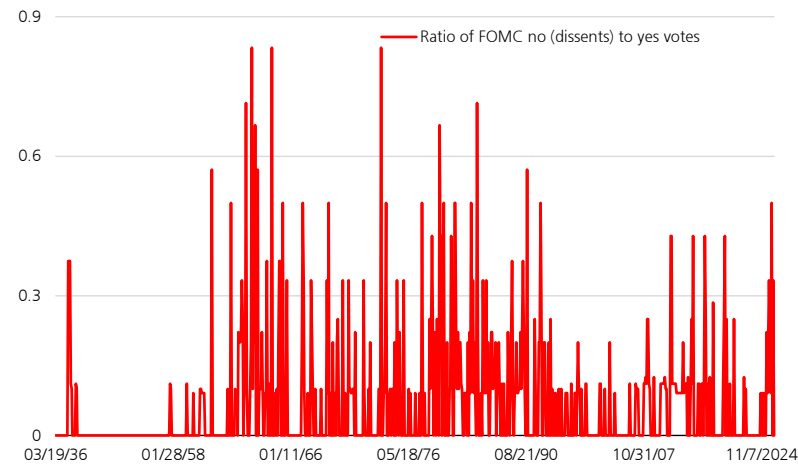
Source: Federal Reserve, BLS, Haver, UBS

# FOMC participants

|                            | 2026                        | 2027                        |
|----------------------------|-----------------------------|-----------------------------|
| Chair of FR Board & FOMC   | Kevin Warsh                 | Kevin Warsh                 |
| Vice Chair of FRB          | Philip Jefferson            | Philip Jefferson            |
| Vice Chair for Supervision | Michelle Bowman             | Michelle Bowman             |
| FRB Governor               | Lisa Cook                   | Lisa Cook                   |
| FRB Governor               | Michael Barr                | Michael Barr                |
| FRB Governor               | Christopher Waller          | Christopher Waller          |
| FRB Governor               | Jerome Powell*              | Jerome Powell*              |
| FOMC Vice Chair            | New York: John Williams     | New York: John Williams     |
| FOMC Member                | Cleveland: Beth Hammack     | Chicago: Austan Goolsbee    |
| FOMC Member                | Philadelphia: Anna Paulson  | Richmond: Thomas Barkin     |
| FOMC Member                | Dallas: Lorie Logan         | Atlanta: ?                  |
| FOMC Member                | Minneapolis: Neel Kashkari  | San F: Mary Daly            |
| Alternate Member           | Chicago: Austan Goolsbee    | Boston: Susan Collins       |
| Alternate Member           | Richmond: Thomas Barkin     | St. Louis: Alberto Musalem  |
| Alternate Member           | Atlanta: ?                  | Kansas City: Jeffrey Schmid |
| Alternate Member           | San F: Mary Daly            | Cleveland: Beth Hammack     |
| Participant                | Boston: Susan Collins       | Philadelphia: Anna Paulson  |
| Participant                | St. Louis: Alberto Musalem  | Dallas: Lorie Logan         |
| Participant                | Kansas City: Jeffrey Schmid | Minneapolis: Neel Kashkari  |

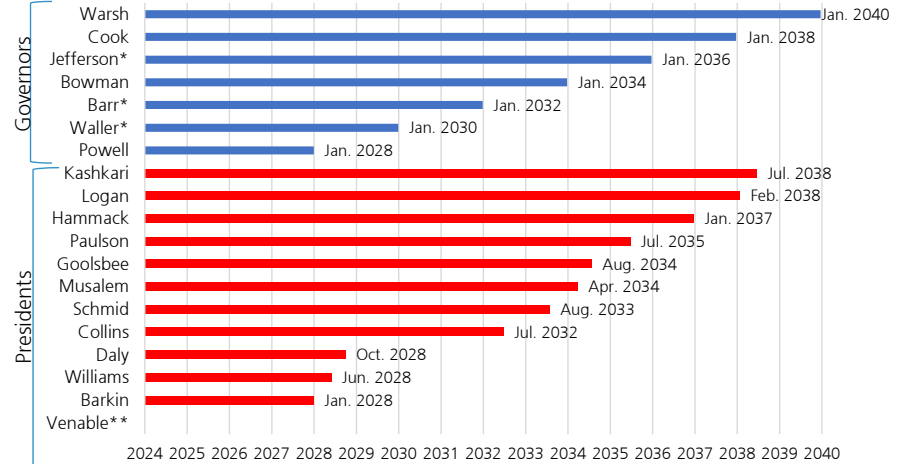
\* Term as Chair ended May 2026, term as Governor ends Jan 2028

## FOMC monetary policy decisions since 1936

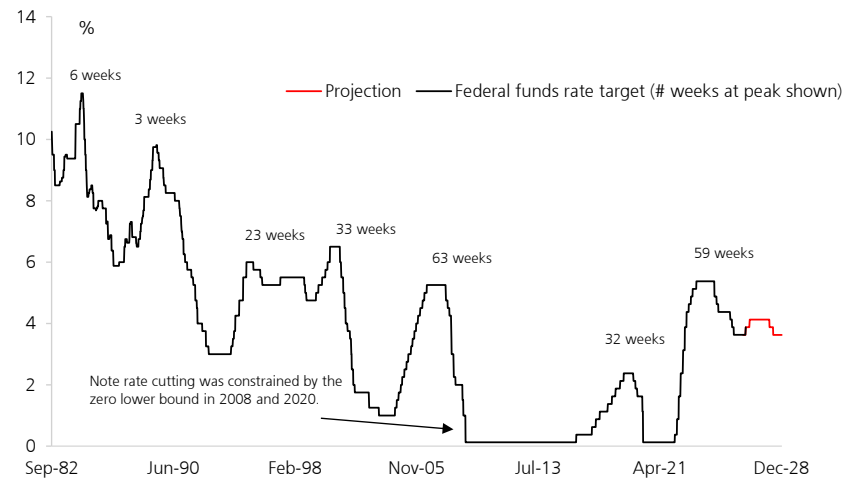


Source: Federal Reserve, Haver, UBS

## FOMC Participants by potential tenure



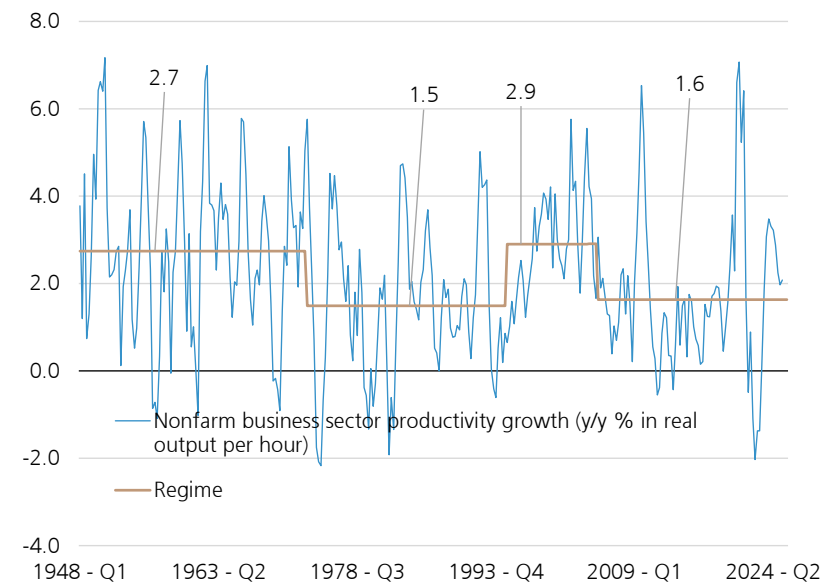
## Fed funds rate forecast



# Structural growth acceleration: aging slows plus AI

- We estimate two sources of acceleration in the coming years. One: waning drag from population aging. Two: we estimate AI to drive a regime shift, accelerating structural productivity growth.
- Immigration is unlikely to be slower in 2027 than in 2025 or 2026. Plus, although the weight of population aging is still negative, the second derivative is turning positive. The weight lessens in the years to come. Not only is structural labor supply poised to accelerate, we think productivity growth is too, thanks to AI.
- We estimate structural US GDP growth is headed to more like 2-1/2% in the coming three to five years, compared to sub-2% FOMC and pre-Covid estimates.

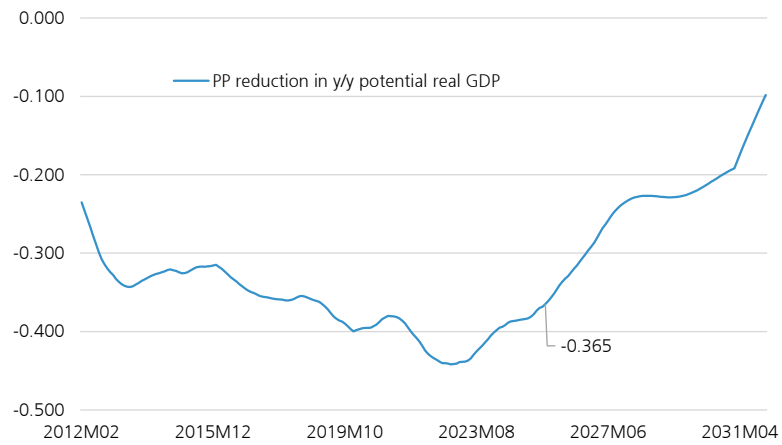
Productivity regimes (% , annualized)



|                      | <u>2018</u> | <u>2019</u> | <u>2020</u> | <u>2021</u> | <u>2022</u> | <u>2023</u> | <u>2024</u> | <u>2025</u> | <u>2026</u> | <u>2027</u> | <u>2028</u> |
|----------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Potential GDP growth | 1.9         | 1.8         | 1.5         | 1.8         | 2.1         | 2.5         | 2.3         | 1.7         | 1.8         | 2.1         | 2.6         |
| Population           | 0.8         | 0.8         | 0.5         | 0.7         | 1.2         | 1.4         | 1.3         | 0.4         | 0.3         | 0.4         | 0.4         |
| LFPR                 | -0.3        | -0.4        | -0.4        | -0.4        | -0.5        | -0.2        | -0.4        | -0.4        | -0.3        | -0.2        | -0.2        |
| Employment rate/AWH  | 0.0         | 0.0         | 0.0         | 0.0         | 0.0         | 0.0         | 0.0         | 0.0         | 0.0         | 0.0         | 0.0         |
| NFB Productivity     | 1.6         | 1.6         | 1.6         | 1.6         | 1.6         | 1.6         | 1.6         | 1.8         | 1.9         | 2.1         | 2.5         |
| GDP/NFB/CES/AWH      | -0.1        | -0.1        | -0.1        | -0.1        | -0.1        | -0.1        | -0.1        | -0.1        | -0.1        | -0.1        | -0.1        |

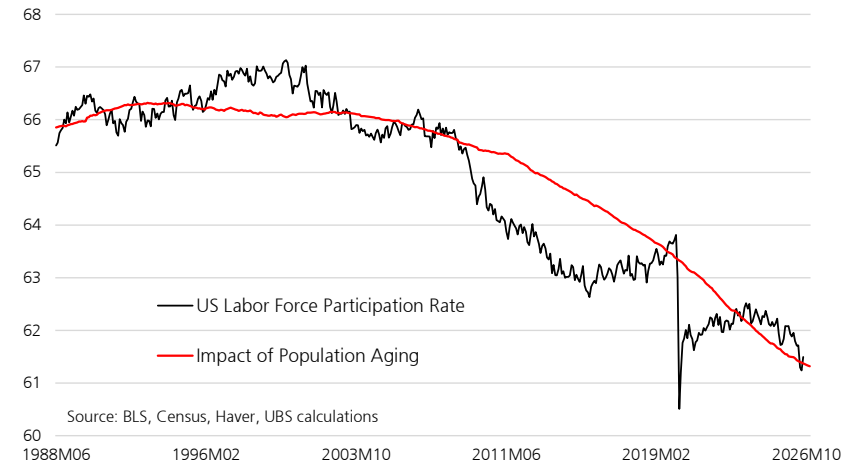
Source: Haver, UBS

Population aging drag on US GDP growth



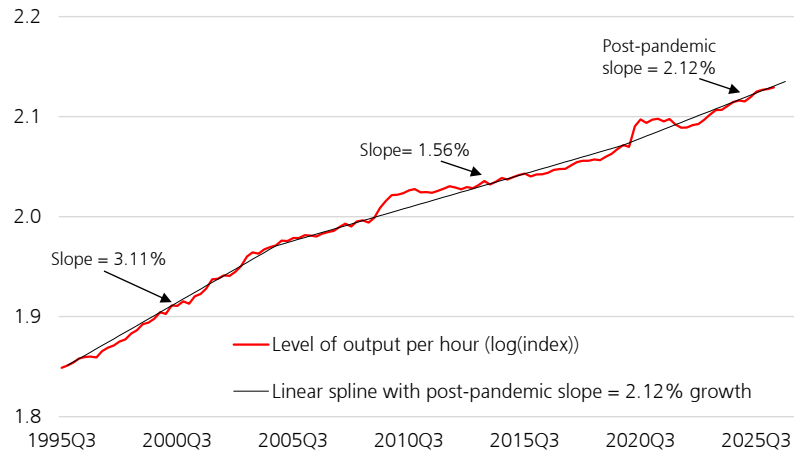
Source: Census, Haver, UBS

Labor Force Participation Rate (% CNIP)

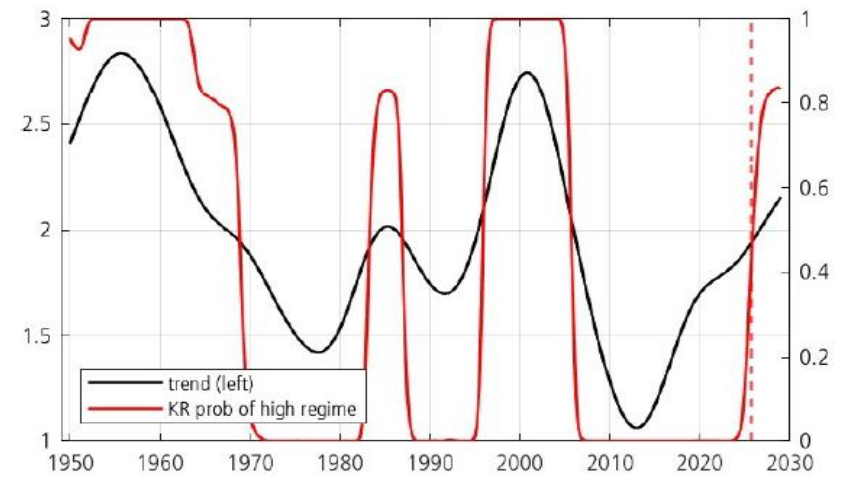


Source: BLS, Haver, UBS

Nonfarm business productivity: (log(index))



Source: BLS, Haver, UBS

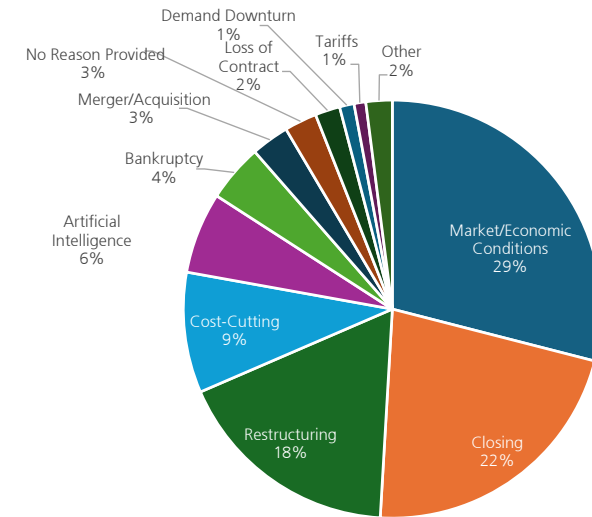


Source: UBS

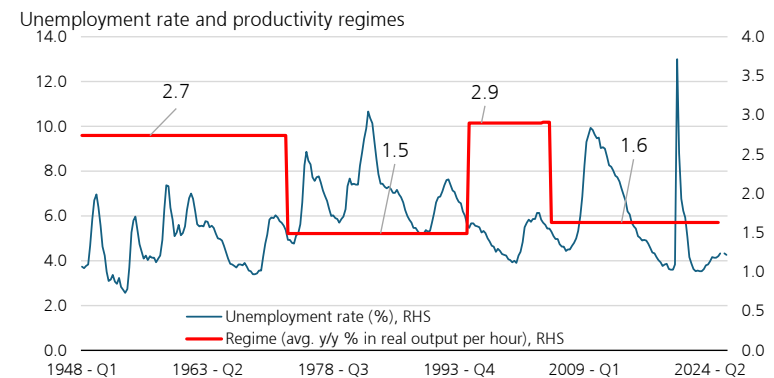
# AI's productivity boost likely to come with more disruption

*We can see AI is having an impact on the labor market, but right now it seems like people see AI in everything. We estimate the impact of AI on the labor market currently to be modest, but we also expect this is still early days for AI disruption.*

- AI in 2025 was cited as the primary reason for 6% of job cuts in announced downsizings. Adoption has been rising quickly as we show on the next page, and less hiring may result outside of just layoffs.
- Incoming data on job destruction that includes firms hiring less, remains quite low, and shows little change from pre-Covid rates. As we show on the next page, job creation and destruction runs between 7 and 8 million jobs per quarter. Those rates should rise if we hit our higher productivity regime that we project.
- Recent college graduate unemployment rates are up 1.7 pp over the year. Is that all AI? Probably not. Labor demand has been weak, including sectors not yet impacted by AI.

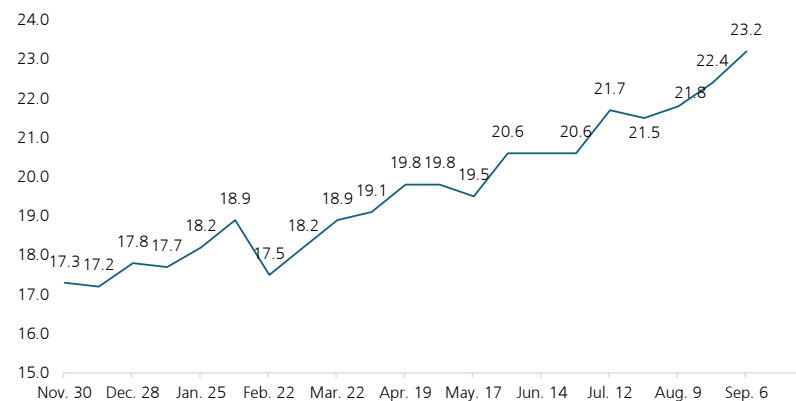


Source: Challenger, Gray & Christmas, Haver, UBS



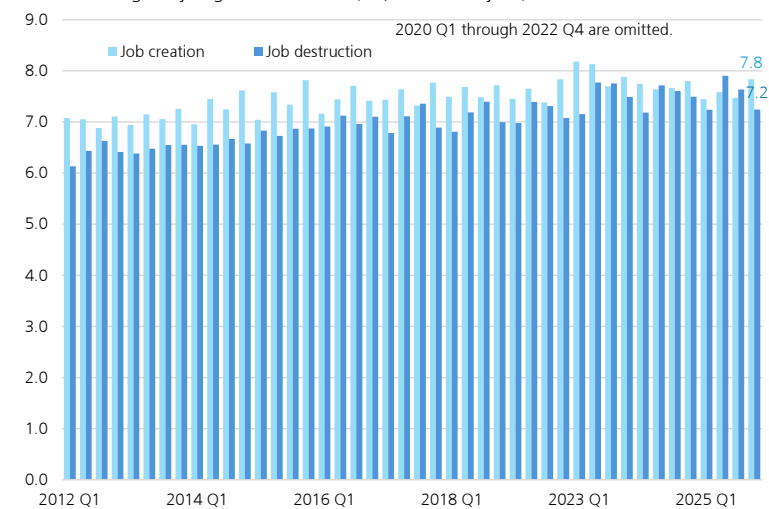
Source: BLS, Haver, UBS

Share of firms responding "yes" to the question, "In the last two weeks, did this business use Artificial Intelligence (AI) in any of its business functions?" (%)



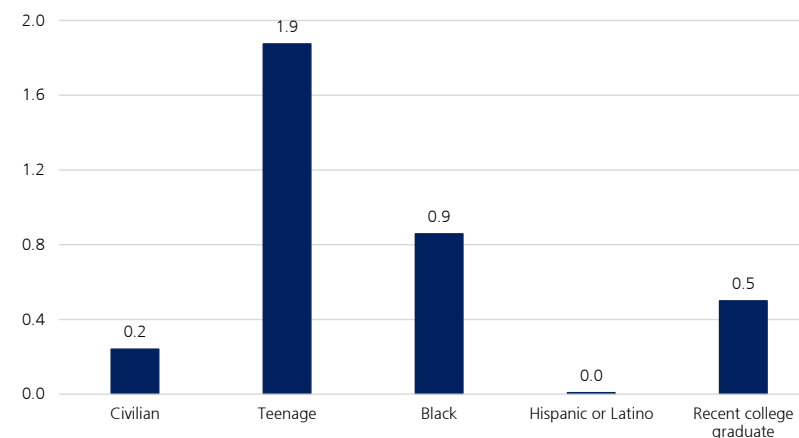
Source: Census Bureau, UBS

Private sector gross job gains and losses (SA, millions of jobs)



Source: BLS, Haver, UBS

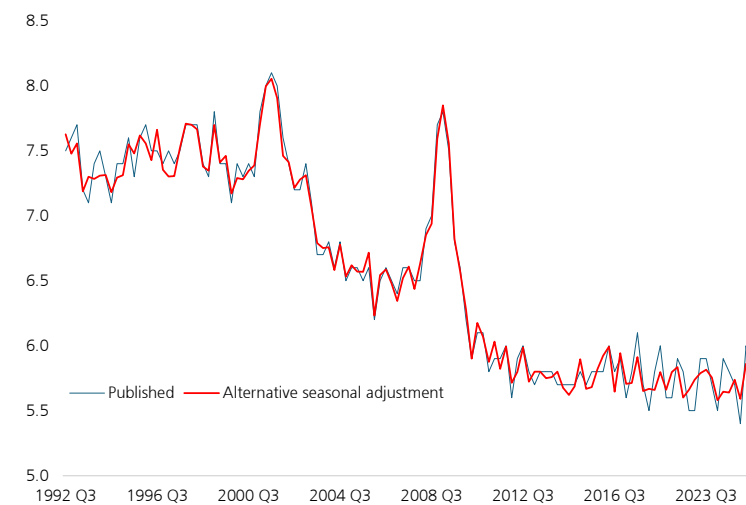
Selected unemployment rates, difference between 2024 avg. and Q2 2026 average level (pp)



Source: BLS, Haver, UBS

Private sector gross job losses as share of employment (SA, %)

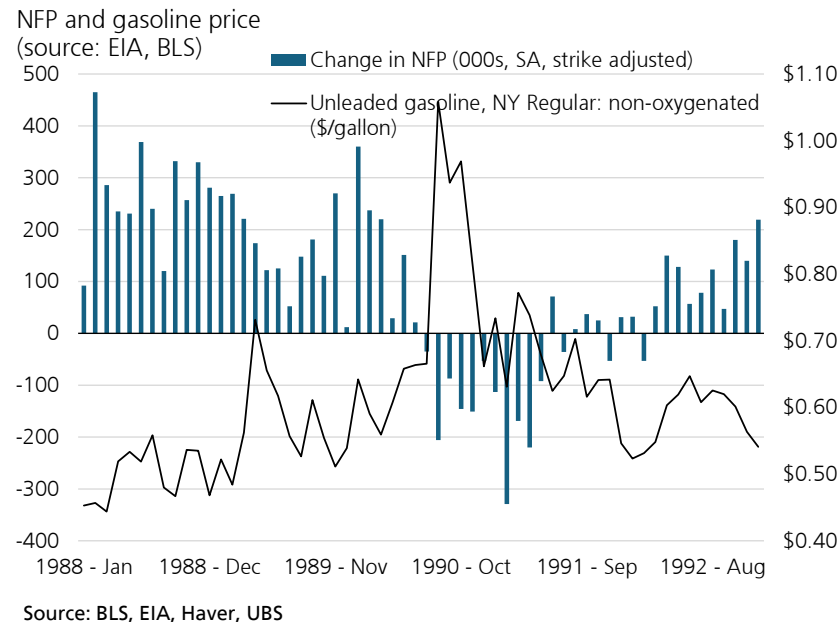
2020 Q1 through 2022 Q4 are omitted.



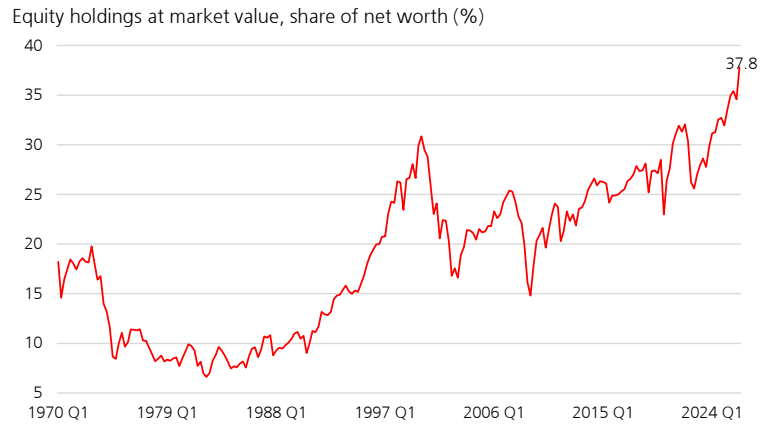
Source: BLS, Haver, UBS

# Risks: energy prices, AI boom, and AI bust

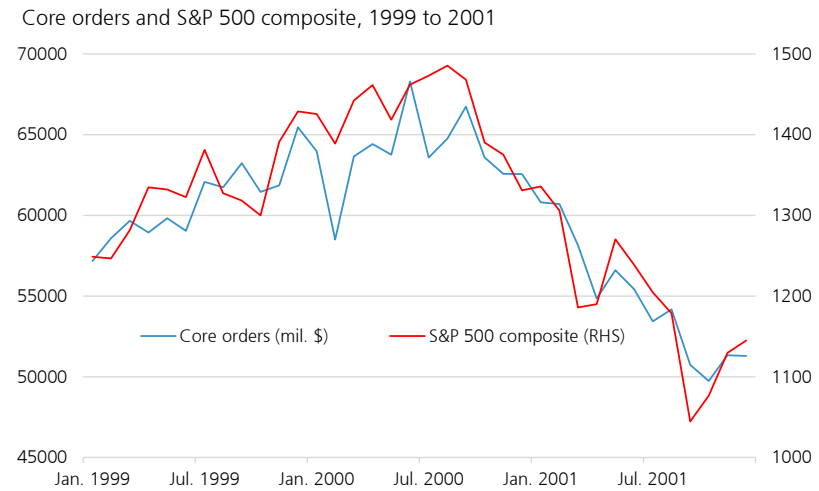
*The AI theme is underpinning the US economy, any faltering here would risk the expansion. On the upside, a greater fiscal boost or even more momentum in tech investment could see stronger growth than our baseline.*



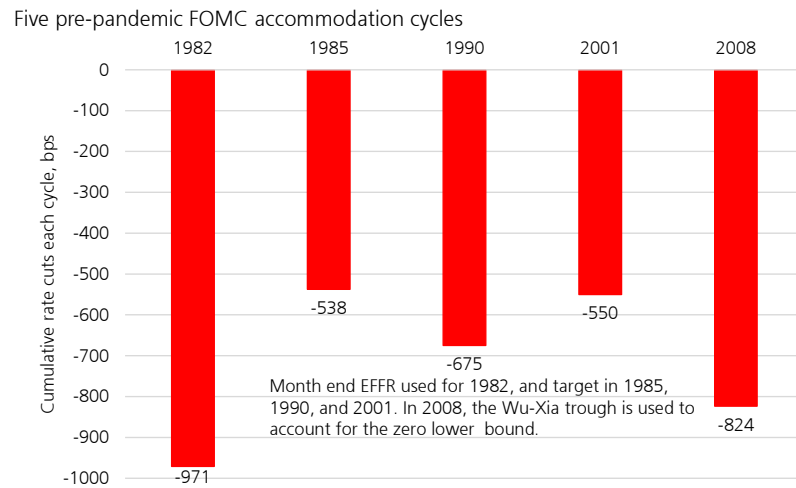
- The cycle is narrowly driven by AI. We modelled an AI bust scenario in which the unemployment rate would rise close to 6%, and inflation would fall below 2.0%. The funds rate returned to the zero lower bound.
- Now, we seem to be piling higher energy costs on a weak labor market, like 1990.
- On the upside, maybe the AI theme drives even more growth. Also, considering the tension between the tariff drag and OBBBA, the stimulus from OBBBA could be greater and longer lasting than we are assuming.
- We are projecting an acceleration in structural productivity growth, which could be followed by higher neutral rates of interest. That should be a slower moving, further out problem, we think.



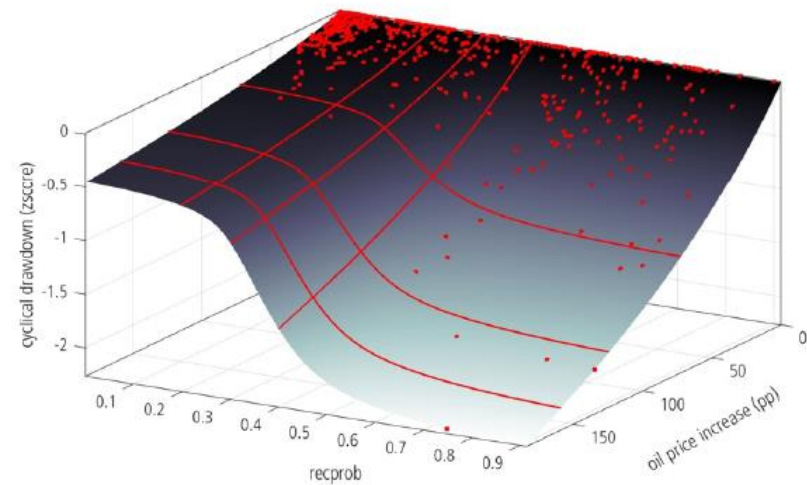
Source: Federal Reserve Board, Haver, UBS



Source: Census, Haver, UBS



Source: Haver, UBS

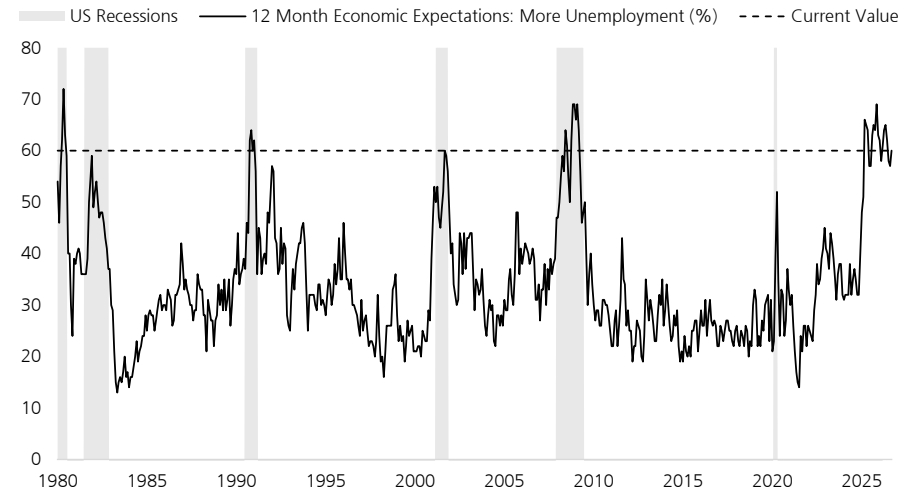


Source: Federal Reserve, Haver, UBS

## Households concerned about the labor market...

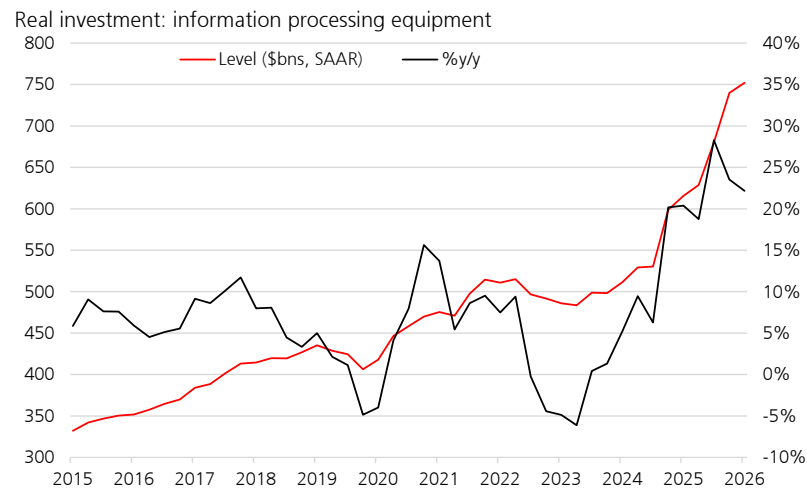


Source: Conference Board, Haver, UBS

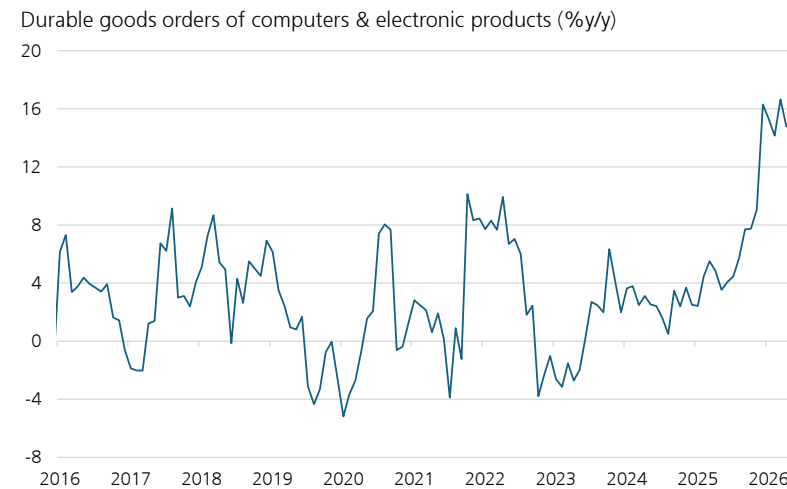


Source: University of Michigan, Haver, UBS

## ... but tech capex is booming



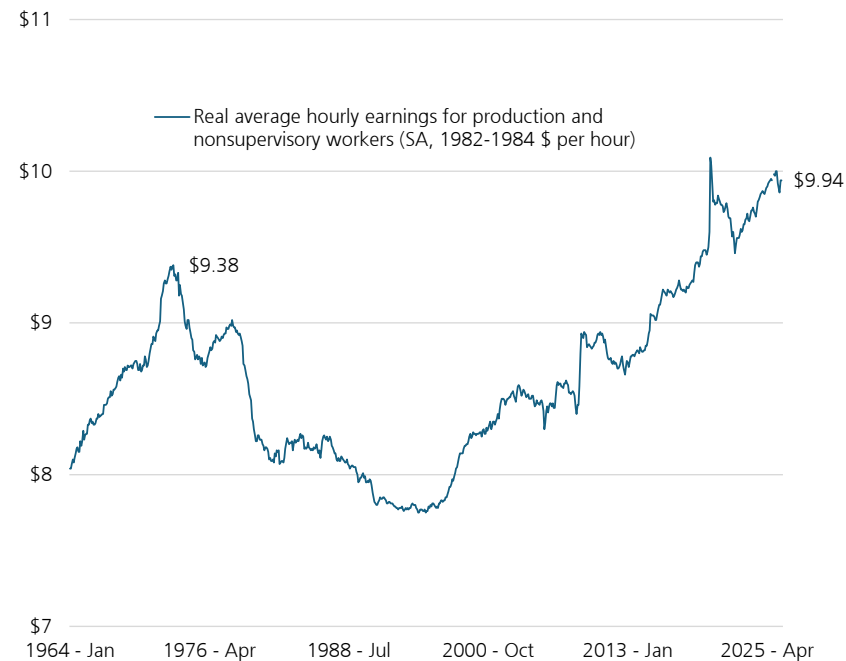
Source: BEA, Haver, UBS



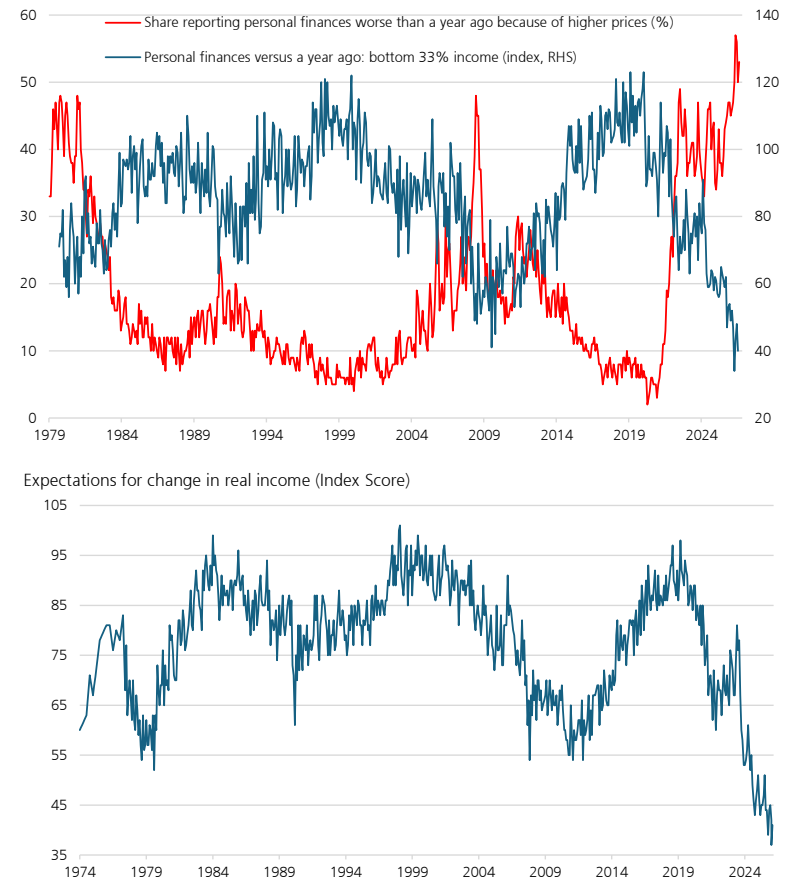
Source: Census Bureau, Haver, UBS

# Mid-term elections ahead: affordability watch

BLS: Real average hourly earnings



Source: BLS, Haver, UBS



Source: University of Michigan, Haver, UBS

# Key Dates

| Month     | Date Day          | Event                                                    |
|-----------|-------------------|----------------------------------------------------------|
| August    | 8/19 Wednesday    | FOMC minutes (Jul meeting)                               |
|           | 8/22 Saturday     | Section 338 tariffs on Canada took effect                |
|           | 8/28 Friday       | Chairman Warsh's Jackson Hole Speech                     |
|           |                   | CES preliminary benchmark revision                       |
| September | 9/4 Friday        | August employment report                                 |
|           | 9/9-10 Wed, Thurs | RNC midterm convention                                   |
|           | 9/15 Tuesday      | FOMC meeting begins (Sep)                                |
|           | 9/16 Wednesday    | FOMC statement & economic projections (Sep)              |
|           | 9/24 Thursday     | President Xi set to visit the US                         |
|           | 9/30 Wednesday    | Annual revisions to GDP/PCE inflation to revise down     |
|           |                   | Fiscal year ends                                         |
| October   |                   | New Medicaid provider tax limits take effect             |
|           | 10/2 Friday       | Employment report (Sep)                                  |
|           | 10/7 Wednesday    | FOMC minutes (Sep meeting)                               |
|           | 10/12-18 Mon-Sun  | IMF Annual Meetings                                      |
|           |                   | IMF finance ministers and central bank governors meeting |
|           | 10/27 Tuesday     | FOMC meeting begins (Oct)                                |
|           | 10/28 Wednesday   | FOMC statement (Oct)                                     |
| November  | 11/3 Tuesday      | US midterm elections                                     |
|           | 11/4 Wednesday    | Quarterly refunding statement (Q4)                       |
|           | 11/6 Friday       | Employment report (Oct)                                  |
|           | 11/18 Wednesday   | FOMC minutes (Oct meeting)                               |
| December  | 12/4 Friday       | Employment report (Nov)                                  |
|           | 12/8 Tuesday      | FOMC meeting begins (Dec)                                |
|           | 12/9 Wednesday    | FOMC statement & economic projections (Dec)              |
|           | 12/11 Friday      | Continuing resolution for FY 2027 funding expires        |
|           | 12/14-15 Mon, Tue | G20 Leaders' Summit                                      |
|           | 12/30 Wednesday   | FOMC minutes (Dec meeting)                               |
|           | 12/31 Thursday    | Medicare physician payment increase expires              |
| 2027      | Q2 estimated      | Debt ceiling binds                                       |
|           | 9/30 Thursday     | State SNAP cost sharing begins                           |

Source: UBS. Debt ceiling estimate from the Bipartisan Policy Center.

# Fed path by meeting

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|      | <u>Meeting</u> | <u>bottom</u> | <u>top</u> | <u>change</u> | <u>mid-point</u> |
|------|----------------|---------------|------------|---------------|------------------|
| 2025 | Jan            | 4.25          | 4.50       | 0.00          | 4.375            |
|      | Mar            | 4.25          | 4.50       | 0.00          | 4.375            |
|      | May            | 4.25          | 4.50       | 0.00          | 4.375            |
|      | Jun            | 4.25          | 4.50       | 0.00          | 4.375            |
|      | July           | 4.25          | 4.50       | 0.00          | 4.375            |
|      | Sept           | 4.00          | 4.25       | -0.25         | 4.125            |
|      | Oct            | 3.75          | 4.00       | -0.25         | 3.875            |
|      | Dec            | 3.50          | 3.75       | -0.25         | 3.625            |
| 2026 | Jan            | 3.50          | 3.75       | 0.00          | 3.625            |
|      | Mar            | 3.50          | 3.75       | 0.00          | 3.625            |
|      | Apr            | 3.50          | 3.75       | 0.00          | 3.625            |
|      | Jun            | 3.50          | 3.75       | 0.00          | 3.625            |
|      | July           | 3.50          | 3.75       | 0.00          | 3.625            |
|      | Sept           | 3.75          | 4.00       | 0.25          | 3.875            |
|      | Oct            | 3.75          | 4.00       | 0.00          | 3.875            |
|      | Dec            | 4.00          | 4.25       | 0.25          | 4.125            |
| 2027 | Jan            | 4.00          | 4.25       | 0.00          | 4.125            |
|      | Mar            | 4.00          | 4.25       | 0.00          | 4.125            |
|      | Apr            | 4.00          | 4.25       | 0.00          | 4.125            |
|      | Jun            | 4.00          | 4.25       | 0.00          | 4.125            |
|      | July           | 4.00          | 4.25       | 0.00          | 4.125            |
|      | Sept           | 4.00          | 4.25       | 0.00          | 4.125            |
|      | Oct            | 4.00          | 4.25       | 0.00          | 4.125            |
|      | Dec            | 4.00          | 4.25       | 0.00          | 4.125            |
| 2028 | Jan            | 4.00          | 4.25       | 0.00          | 4.125            |
|      | Mar            | 3.75          | 4.00       | -0.25         | 3.875            |
|      | Apr            | 3.75          | 4.00       | 0.00          | 3.875            |
|      | Jun            | 3.50          | 3.75       | -0.25         | 3.625            |
|      | July           | 3.50          | 3.75       | 0.00          | 3.625            |
|      | Sept           | 3.50          | 3.75       | 0.00          | 3.625            |
|      | Nov            | 3.50          | 3.75       | 0.00          | 3.625            |
|      | Dec            | 3.50          | 3.75       | 0.00          | 3.625            |

Source: Federal Reserve, Haver, UBS. July 2026 onward are UBS projections.

# What next? More bumps in the road?

*The expansion rests on AI and the related wealth gains. We are now passing peak fiscal support to growth, and monetary policy tightening and fiscal policy turning to drag means the two will be working in the same direction for the first time in years. In contrast, we are past peak tariff impact on inflation and potentially peak energy shock inflation. Will energy prices fall back, or supply constraints rise? The labor market has been resilient, but details suggest more vulnerability than appreciated. If the AI investment or support for equities falters, the expansion itself would be at risk. The FOMC now appears headed toward further restraining the ex-AI economy. As a base case, we expect the FOMC to raise the Fed funds rate twice in 2026. We doubt that dampens the AI investment momentum, within which much of the investment is imported and not payments to domestic workers or income. Rate hikes will work to slow the rest of the economy. Plus: Can the geopolitical disruptions, policy uncertainty and volatility get out of the way?*

| <u>Variable</u>    | <u>2023</u> | <u>2024</u> | <u>2025</u> | <u>2026</u> | <u>2027</u> | <u>2028</u> |
|--------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Change in real GDP | 3.4         | 2.4         | 2.0         | 2.2         | 1.9         | 2.6         |
| Unemployment rate  | 3.8         | 4.2         | 4.5         | 4.2         | 4.5         | 4.4         |
| PCE inflation      | 2.9         | 2.6         | 2.8         | 3.6         | 2.0         | 1.9         |
| Core PCE inflation | 3.3         | 3.0         | 2.8         | 3.1         | 2.4         | 2.1         |
| Federal funds rate | 5.4         | 4.4         | 3.6         | 4.1         | 4.1         | 3.6         |

Note: Q4/Q4 % basis except the unemployment rate which is a Q4 quarterly average. The federal funds rate reflects the mid-point of the target range. Years after 2025 are forecasts. Inflation reflects post-Sep. 30 NIPA revisions.

Source: BEA, BLS, Haver, UBS

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